Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

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Abstract

What a recorded field is worth to a prediction is reported as one number, but that number is a functional of rarely stated choices: baseline, pipeline, target, split, decision time, register quality, metric and operating point. We define incremental predictive performance as a *specification surface* $V_s(f)$ over a declared design space under a declared measure, and give four objects for reporting one: an exact functional-ANOVA decomposition of its variance; simultaneous max-t bands over the *whole* surface a claim ranges over, from a pipeline-refitting moving-block bootstrap; *resolution regions* with a robustness index; and *specification regret*. On 19 log–target pairs from 13 public event logs, over an audited 29,040-cell space, the baseline alone moves the increment by 0.073 AUC at the median pair, and a one-number report misstates the sign of 7.5% of resolved cells — 3.5% on the 8 maintained registers. Of the 13 pairs resolving any sign, 3 resolve both. On 40 of 118 models a threshold is not a cost ratio at all. In a case study on 45,455 bank incidents, a register's value turns on the decision time and its sign on the target.

Keywords: specification surface, incremental predictive performance, predictive process monitoring, simultaneous inference, sensitivity analysis, configuration management database

Highlights

- A field's incremental predictive performance is a surface, not one number

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- An exact variance decomposition says which analyst choice moves the answer
- Coverage-calibrated bands say where the sign of that surface is resolved
- Sign misstated for 7.5% of resolved cells, 3.5% on ITSM
- A register's worth turns on the decision time and on which target is set

1. Introduction

A claim that a recorded field is worth something is a comparison. Its second arm — performance *without* the field — is a choice the analyst makes, and so are the learner that makes the comparison, the encoding it uses, the outcome it predicts, the way the data are split in time, whether the field is available at the moment the decision is taken, how completely the register behind the field is populated, the metric the difference is read in, and the operating point at which that metric is evaluated. Each is routinely left implicit, and the result is reported as one number.

This paper's position is not that such a number depends on those choices — that is known, and Section 2 says by whom — but that the dependence is large enough, on real decisions, to change the sign of the answer; that the right object to report is therefore a surface over a *declared* design space rather than a point; and that a surface admits estimation and inference of its own.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) \;=\; m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}),\, D_{\text{test}}) \;-\; m_\theta(A_s(D_{\text{train}}, B),\, D_{\text{test}})\,, \quad (1)$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q, m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one.

We say *incremental predictive performance* and not *value*: (1) measures how much better a model predicts, not what the field costs to acquire or maintain, whether acting on the prediction changes the outcome, or what an organisation would pay for either. Section 8 converts the increment into net

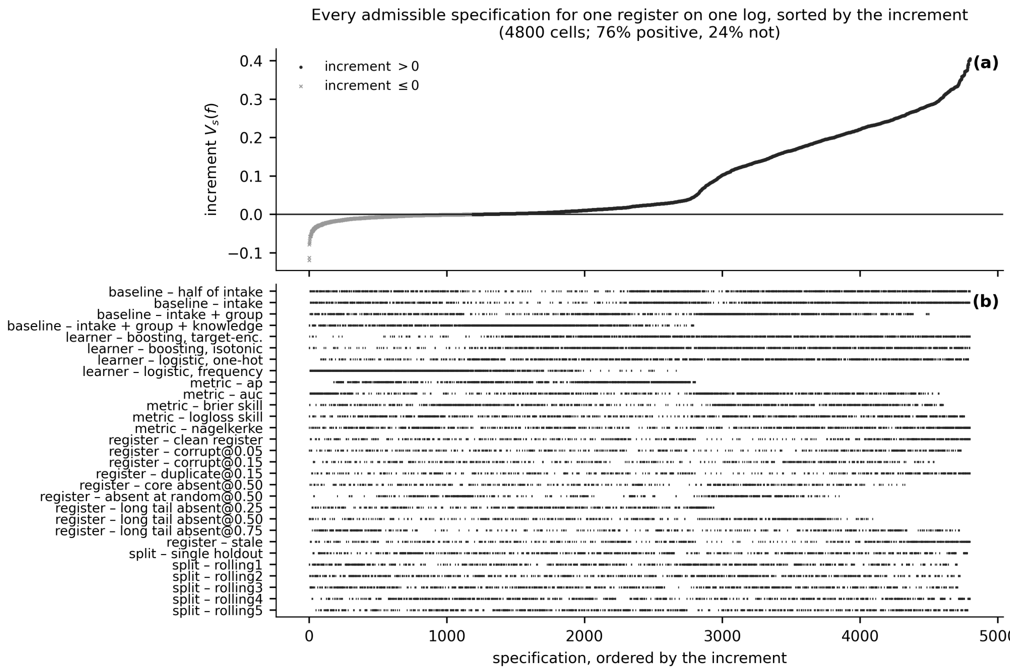

Figure 1: The specification curve for the configuration item on BPIC14, at the registered handover target. **(a)** Each point is one admissible specification — a complete assignment to the pipeline, the baseline rung, the register-quality condition, the split and the metric — sorted by the increment. **(b)** Which level of which axis each specification occupies. The intercept-only rung is excluded. Every point is a point estimate; the simultaneous bands that decide the region labels are in Table 5. **The five scalar instruments are plotted on one axis and their increments are not on a common scale** (Section 4.7); what the panel supports is the *sign* statement, which is unit-free, and not a comparison of magnitudes across instruments. The panel shows the increment rather than the reduction, so a cell where the register is worth almost nothing looks the same as one where the free field absorbs almost everything.

benefit at a declared exchange rate, which is as close to economic value as this evidence gets.

Contributions. Multiverse and specification-curve analysis, functional-ANOVA and Sobol indices, the moving-block bootstrap, max-t and multiplier bootstraps, decision curve analysis, calibration assessment, and feature and data valuation are all established, and none of them is invented here; Section 2 attributes each and states the gap they leave. What follows is what this paper adds.

1. **A new formalisation: the specification surface and a complete estimand.** Section 3 defines $\mathcal{S}$, V_s, and the relative reduction R, states the conditions under which R is a quantity at all, and proves two results. R is *not identified* by the data, the feature and the baselines: there is no function carrying the reduction under one metric to the reduction under another (Proposition 1). And, relative to a *declared family* of recalibrations, R survives every member of that family exactly when each member acts *affinely* on the instrument (Proposition 2) — which within the affine family is strictly weaker than rank-basedness, and outside it is not (Section 3.5). The proposition also converts a search into a proof: one violating triple settles an instrument for the family it was found in.
2. **New reporting objects: two adaptations and two new.** Section 4 supplies the estimation and inference layer. Four constructions are taken from the literature and two of them are *adapted*: (i) an exact, orthogonal functional-ANOVA decomposition of $\mathrm{Var}_s[V_s(f)]$ into first-order and total indices per axis, taken here under a *declared measure* over specifications rather than under whichever cells happened to be enumerated; (ii) a moving-block bootstrap that resamples the training half, *refits the entire pipeline*, and evaluates on a resampled test half, so the interval contains training-sample, model-selection, encoder and calibrator variability; (iii) simultaneous max-t bands whose family, for a surface-level claim, is the *whole surface* that claim ranges over and not a cell of it, with the critical value estimated by a multiplier bootstrap so that its precision is not limited by the refitting budget; and (iv) Fieller intervals for R, which report an unbounded or exclusive confidence set when the denominator is weak instead of a bounded interval that is not one.

   Two further objects are new as far as we are aware. A cell is *beneficial*, *harmful* or *unresolved* under the simultaneous band, a surface carries one of six *resolution region* labels, and the robustness index $\rho \in [-1, 1]$ summarises it; $\rho = 1$ is the only state in which one positive number is a safe summary. And *specification regret* measures what one-number reporting costs — the share of admissible specifications whose sign disagrees with the reported cell, and the performance forgone or destroyed by acting on it, computed *inside one instrument at a time*. Lemma 1 identifies the decision-optimal collapse, the weighted mean increment, and Section 6.8 measures four rules against it *out of sample*, on held-out specifications, held-out time and held-out logs.

3. **New empirical evidence: a registered multi-log benchmark.** The surface is computed through one frozen pipeline on *every* log–target pair a pre-registered protocol admits — 19 of them, not only the ones where the answer resolves — over an audited denominator, with calibration, unseen-category rates and rolling-origin validation reported for each, and every corpus statistic is reported twice: on all 19 pairs and on the 8 where the register is a maintained one and the construct is what the paper is about (Section 6.6). Section 6.7 runs specification-curve analysis as practised on the same surfaces and reports where its verdict and the region label part company, which is the concrete statement of what the extra machinery buys.
4. **A new operational case: a register at three decision times.** Section 7 shows that a decision time fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — estimates a surface at each of three moments in the service-desk process, and separates the three mechanisms. It also reconciles the two values this log has been reported to give, and finds that the difference is the *target*: over the cohort × target design the target carries 99.3% of the variance and the cohort 0.0%.

What this paper is not. This is a retrospective study on public benchmarks, offered as an evaluation methodology. It is not a procurement conclusion about configuration management databases, not a systematic review of reporting practice, and not a report of a deployed system; Section 11 says what each of those would have required.

The supplementary document. The protocol and its amendments, the proofs, the constructions of the four reporting objects, the correction register, the results this article summarises and the tables it does not print are in a supplementary document whose sections, tables and figures carry an S. A reference here to Section S3 or Table S12 is a reference to that document; the article is self-contained without it.

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.5	4	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	5	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.6	3	a function of the band; no further refit
Specification regret	what choosing one specification costs against the best one, on a scale a decision can use	4.7	S11	exact; no refit, on the surface already computed

Table 1: The four reporting objects this paper supplies, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

2. Related work

Variable importance and incremental value. That a feature's apparent importance depends on what else is in the model is the standard motivation for conditional and permutation-based importance [36, 13], for Shapley-value attributions [23, 8], and for nonparametric formulations of importance as a population quantity [45], which average over subsets precisely because a single conditioning set is arbitrary. The clinical-prediction literature has made the same point about the *incremental* value of a marker for two decades: the increase in the area under the ROC curve from adding a predictor depends on how good the existing model is [28, 7], is a poor guide to clinical usefulness [7, 40], and is routinely misread [20]. What that literature establishes, and what we take as given, is that the baseline and the metric are choices.

What it does not supply is an object that reports the whole dependence, an estimator for it, or inference over it.

Decision-analytic evaluation. Decision curve analysis [42, 43] makes the operating point explicit by reading net benefit across an exchange rate, and is the reason the operating point is an axis here rather than a footnote. Guidance on reading such a curve is well developed [41]; what is less settled is what a claim about a *band* of thresholds requires, which is why the bands in Section 8 are simultaneous rather than pointwise.

Specification sensitivity. That analytic conclusions move with defensible analytic choices is the subject of multiverse analysis [34], specification-curve analysis [32], vibration of effects [27] and, in machine learning evaluation, of work on underspecification [9] and on the fragility of benchmark rankings [3]. Our design space and our figures are recognisably in that tradition. What this paper adds to it is a decomposition of the surface's variance into axis contributions, simultaneous inference over the surface rather than a distribution of point estimates, region labels with an explicit decision rule, and a regret benchmark against the conventional report. Specification-curve analysis, as usually practised, displays the curve and applies a permutation test to its median; it does not tell a reader which sub-region of the design space supports a sign.

Data valuation. Shapley-based data valuation [14, 17] values *instances* against a fixed learner and metric; the axes we vary are exactly the ones held fixed there. The two are complementary: a data value is itself a point on a specification surface.

Which axis a study varies. Some studies do vary one axis deliberately: the metric axis in effort-aware defect prediction [4], the encoding axis in remaining-time prediction [22], the inter-case feature block in predictive process monitoring [31]. What is missing is an object that holds several axes at once and says what their joint variation implies for a sign.

Prediction on incident event logs. The empirical setting is predictive process monitoring, surveyed and benchmarked across real logs [38, 39] on data of the kind process mining formalises [1]. We predict at case creation from creation-time attributes with no prefix, which is tabular classification on a process log rather than prefix-based monitoring, and that difference bounds what the corpus result transfers to. The benchmark protocols in that literature fix a learner, an encoding and a metric per experiment, and this paper measures the sensitivity of a conclusion to *those* fixings. It does not measure sensitivity to **prefix length, prefix bucketing or sequence encoding**, which are the axes that literature varies most, because none of them exists at a single prediction point; Section 11 carries the transfer as an assumption rather than a result. Configuration management databases and their data quality are treated in the IT service management literature as an operational problem [24]; we are not aware of an evaluation of one against a named prediction task with the design space held open.

Data quality as an axis. The effect of data quality on model performance has been studied directly on tabular data [26] and, in process mining, for low-quality activity labels [6]. Those studies vary quality with the feature set fixed; we vary quality *as one axis of the same surface* on which the feature set and the encoding are others, which is what makes the interaction between them visible — Section 7.4 reports a quality defect that is exactly invisible under one encoder and not under another. Leakage, which is availability misspecified, has its own literature [19, 18, 44], and the decision time of Section 7.2 makes it a declared choice rather than a mistake.

The novelty gap, precisely. Every component used here is established and none is claimed. What none of it supplies is the object this paper reports, and the gap is in four places. *The estimand is not declared completely*: the axes are discussed one at a time, and the decision time and the register's quality are treated as preliminaries rather than as arguments. *The measure is left implicit*: a specification curve is summarised uniformly over whatever cells were enumerated, which is a choice that moves the answer (Section 4.5). *Inference is pointwise or absent*: a specification curve is a distribution of point estimates and a permutation test on its median, which does not license a statement about a sub-region. *Nothing measures what the conventional report costs*: the alternative to one number is stated as a display, not as a decision rule with a loss. Of the objects that close those four, the variance decomposition and the simultaneous band are adaptations, while *resolution regions* with a robustness index and *specification regret* are, as far as we are aware, new, and Section 6.7 reports where a specification curve's verdict and a region label part company so that the difference is measured rather than asserted.

The contribution is deliberately *not* stated as "people usually report one number". It is stated as: a claim of this form is a functional of a design, the functional can be estimated and its uncertainty quantified, and on this corpus it is far from constant.

3. The specification surface

3.1. The design space

Definition 1 (Design space)**.** *A* design space $\mathcal{S}$ *for a feature-value claim is a finite set of specifications, each a complete assignment to the axes*

$$s = \Big(\underbrace{A}_{\textit{learner}}, \underbrace{E}_{\textit{encoding}}, \underbrace{Y}_{\textit{target}}, \underbrace{\Pi}_{\textit{split}}, \underbrace{\tau}_{\textit{decision time}}, \underbrace{q}_{\textit{register quality}}, \underbrace{B}_{\textit{baseline}}, \underbrace{m}_{\textit{metric}}, \underbrace{\theta}_{\textit{operating point}} \Big),$$

together with the cohort and eligibility rule that fix D. A claim about a feature is well posed *only relative to a declared $\mathcal{S}$.*

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level that is deliberately excluded with the reason. Table 2 is that declaration for this study, and it is generated from the code that walks the space rather than typed beside it. A reader can multiply its level counts and obtain the cell counts reported in Section 6; that this multiplication works is the first check the verification harness runs.

On the measure. A design space is a set; a decomposition, a weighted mean or a robustness index needs a *distribution* on it, and "uniform over the cells I enumerated" is not neutral — it moves if a level is duplicated, if five folds are written as five levels rather than one, or if a grid is refined. This paper separates the weight on an axis from the weights within it, declares three product measures and an envelope over a fourth (Section 4.5), and reports every weighted quantity under all of them.

Two of these axes deserve comment because they are usually not treated as design choices at all.

The **decision time** τ fixes which fields exist when the prediction is made. It is not the same as the split rule and it is not a data-cleaning detail: moving τ later in a process admits fields written in between, and Section 7 is a case

axis	levels	measure	exclusions
learner	2-4	equal / ref. / conc.	none
encoding	2-3	set by the learner	none
split	6	equal / ref. / conc.	none
decision time (availability)	1-2	never averaged	two times on the primary log, one elsewhere
register quality x severity	6-10	equal / ref. / conc.	none
baseline (ladder rung)	4-5	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3-4	equal / ref. / conc.	none
scalar instrument	5	never averaged across	none
operating point (net benefit)	31	declared threshold dist.	outside the declared range

Table 2: The declared design space. One row per axis: the number of levels (a range where it differs by log), the measure the paper places on those levels, and every level excluded by declaration with its reason. Generated from the code that walks the space, so the level counts multiply to the cell counts of Table S7.

in which that single move takes a field's increment from resolvably positive to indistinguishable from zero.

The **register quality** q acts on f rather than on the comparison. A field that is a lookup into a maintained register — a configuration item, a customer master record, a product catalogue — inherits that register's completeness and accuracy. We implement six mechanisms besides the clean field, each parameterised by the training half alone: population loss with the long tail absent, with the core absent, and at random; identity error, in which a share of rows carry a wrong value drawn from the training alphabet; reconciliation failure, in which a share of identities exist twice; and staleness, in which an identity is present only if the register had already seen it by the split point. The first three model coverage, the next two accuracy, the last discovery; the taxonomy follows the event-log imperfection patterns of Suriadi et al. [37] where they apply.

Real registers do not fail at random, so three *further* mechanisms make the missingness depend on something: on *time*, where a row's probability of carrying an identity rises with its position in the log; on *content*, where coverage depends on the item's recorded type through a map estimated on the training half; and on a *propensity* built from the intake block, which makes missingness correlated with the outcome without ever reading a test outcome. Every mechanism is swept over 11 severities and every stochastic one repeated over 20 seeds, and each is verified by execution to be a function of the training half alone — perturbing the test half and re-degrading must leave the training half's degraded values identical, which `results/s01_trainonly.csv` records over 122 checks.

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test}, so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. That is not the quantity this paper reports intervals for, and the distinction has to be made before Section 4 can say what its bootstrap targets.

Definition 2 (The estimand)**.** *Let the cases of a log be a sample from a population P, and let the specification s fix the fitting procedure A_s, the cohort and eligibility rule, the split rule Π, the decision time, the register-quality mechanism, the baseline and the instrument. Write $A_s(P)$ for the limit of the fitted scoring rule as the sample grows under P with s held fixed. The* estimand at *s is*

$$\vartheta_s(f) \;=\; m_\theta\Big(A_s(P, B \cup \{f_q\}),\, P\Big) \;-\; m_\theta\Big(A_s(P, B),\, P\Big),$$

and $V_s(f)$ of (1) *is its plug-in estimate on the log.*

Three things follow and are used throughout. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away: ϑ_s is what this analyst's procedure converges to, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it, and why the split is an axis. Under misspecification ϑ_s is not what the register is worth in the world — the simulation of Section 10 separates the two and measures the gap, calling them V_{limit} and V_{oracle}. And every cell of every figure and table in this paper is an *estimate* of ϑ_s; "resolved" in Definition 3 means that the band excludes zero for ϑ_s, not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D, the surface is the map $s \mapsto V_s(f)$ of (1), and a conventional report is one cell of it. The paper's claim is that a single cell is not a summary of the surface in any useful sense: Section 6 shows that the sign of V_s disagrees across cells on most log–target pairs we study, and Section 6.5 measures how often. There is one state in which a scalar is defensible: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive number is a lower bound on a robust conclusion. Section 4.6 makes that a definition.

3.4. The relative reduction, and its pathologies

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) \;=\; V_s(f \mid B_0) - V_s(f \mid B_1), \quad R_s(f) \;=\; 1 - \frac{V_s(f \mid B_1)}{V_s(f \mid B_0)} \;=\; \frac{D_s(f)}{V_s(f \mid B_0)}. \tag{2}$$

D is an absolute difference in the metric's own units. R is a ratio and inherits every pathology of one: the denominator can be small, zero or negative; R is not comparable across metrics, because Proposition 1 says the two are not functions of each other; and R can exceed 1 or fall below 0, so reading it as "the share absorbed" is a category error whenever it leaves $[0, 1]$.

We therefore make D and the two increments primary throughout and treat R as a secondary descriptive statistic subject to three rules, declared before any R in this paper was computed:

1. R is reported only where $V_s(f \mid B_0)$ is resolvably positive under the *simultaneous* band, not merely positive as a point estimate;
2. its interval is a Fieller set and its *kind* — bounded, unbounded, or the complement of an interval — is printed beside it. A confidence set for a ratio is the solution set of a quadratic [12] and is a bounded interval only when that quadratic's leading coefficient is positive, which is to say only when the denominator is resolvably away from zero; otherwise it is unbounded or is the *complement* of an interval, and no bounded interval summarises it (Supplement S3). Of the 10,584 reductions this paper computes, 5,282 have a non-bounded Fieller set, and a percentile interval reports a bounded one for every single one of them;
3. the magnitude of R is never compared with the spread of R across metrics as though both lived on an unproblematic common scale.

3.5. Two propositions

Throughout, a *configuration* is a quadruple of score vectors on a fixed labelled test set, $\mathcal{C} = (\sigma_{B_0}, \sigma_{B_0 \cup f}, \sigma_{B_1}, \sigma_{B_1 \cup f})$ with $B_0 \subset B_1$, and $V_m(B)$, $R_m(\mathcal{C})$ are as in (1) and (2). The metric conventions the constructions below depend on — the tie rule for ROC AUC, the step-wise definition of average precision, the selection rule for net benefit — are fixed and listed in `results/s07_conventions.csv`.

Proposition 1 (Metric non-identifiability of the reduction)**.** *There is no function $\varphi : \mathbb{R} \to \mathbb{R}$ with $R_{\mathrm{AP}}(\mathcal{C}) = \varphi\Big(R_{\mathrm{AUC}}(\mathcal{C})\Big)$ for every configuration $\mathcal{C}$ whose denominators are positive under both metrics. Consequently the reduction is not identified by the data, the feature and the two baselines: the metric is an argument of the estimand.*

The construction is in Supplement S2.

It establishes non-identifiability and not unboundedness: R is built over a bounded interval in any fixed configuration, so the reading is that no function carries one metric's reduction to another's. In the two configurations of `results/s29_prop1.csv` the AUC reduction is identical to 2.2×10^{-16} while the average-precision reduction differs by 0.417.

The second proposition concerns what a monotone recalibration of the scoring system does to the reduction. It is stated *relative to a declared family of recalibrations*, which is both what its proof reaches and what an analyst can act on: nobody recalibrates by an arbitrary monotone map, and a reader who has declared that they will consider Platt scaling, or isotonic regression, or any affine rescaling, is asking about that family and not about all of them.

Assumption 1 (Richness)**.** Every quadruple of score vectors on the fixed labelled test set is realisable as a configuration. This is the assumption that no restriction is placed on the model class; it is what makes "for every configuration" quantify over a set with more than three elements, and it fails only if the analyst commits in advance to a family of scoring rules narrow enough to constrain the four arms jointly.

Richness is used only in the direction (i) $\Rightarrow$ (ii), so under a model class too narrow to realise the separating configuration an instrument can be invariant on *that class* without acting affinely; the direction that matters for practice, (ii) $\Rightarrow$ (i), needs no richness at all. A reader working inside a restricted family should therefore read the proposition as a sufficient condition and establish non-invariance on their own class by exhibiting a configuration in it, which is what the certificates below do.

Proposition 2 (What survives a declared family of recalibrations)**.** *Let Ψ be a family of strictly increasing continuous maps containing the identity, each applied* simultaneously *to all four score vectors of a configuration — one recalibration of the scoring system, not a different one per arm. Call m* Ψ-invariant *if $m(\psi(\sigma), y) = m(\sigma, y)$ for every $\psi \in \Psi$ and every (σ, y). Under Assumption 1, the following are equivalent.*

(i) $R_m(\psi\mathcal{C}) = R_m(\mathcal{C})$ *for every configuration* $\mathcal{C}$ *with nonzero denominator and every* $\psi \in \Psi$.

(ii) For every $\psi \in \Psi$ *there are constants* $A(\psi) \neq 0$ *and* $B(\psi)$ *with* $m(\psi(\sigma), y) = A(\psi)\, m(\sigma, y) + B(\psi)$ *for every* σ.

The proof is in Supplement S2; it is three lines and is moved only for length. Taking Ψ to be every strictly increasing continuous map recovers the unrestricted statement, and taking it to be the recalibrations an analyst has declared is the version that can be checked.

Within a declared family, invariance is strictly weaker than rank-basedness. A rank-based instrument satisfies (ii) with $A \equiv 1$, $B \equiv 0$ for every Ψ, hence satisfies (i); the converse fails for some Ψ. Let Ψ_{aff} be the affine rescalings and m the gap between the mean score on the positives and the mean score on the negatives. It is not rank-based — an affine recalibration changes it, by up to 0.080 over the configurations in `results/s29_wider.csv` — yet ψ acts affinely on it with $A(\psi)$ the slope, so its reduction is Ψ_{aff}-invariant to 1.9×10^{-14}.

The separation is family-relative. The same witness does *not* separate at Ψ = all monotone maps: over 5 non-affine recalibration families the class-mean gap's reduction shifts by up to 0.638, against 1.9×10^{-14} inside Ψ_{aff}. Whether any instrument satisfies (i) unrestrictedly without being rank-based is a question this paper does not settle and does not need to: an analyst declares the recalibrations they will consider, and the proposition answers for those.

What this buys is that a search becomes a proof. To show that an instrument's reduction is *not* invariant it suffices to exhibit one ψ and one triple (a, b, c) violating (1) — a *certificate* that ψ does not act affinely on m — because the proposition then rules out invariance for every configuration. `results/s29_certificates.csv` carries 18 of them, one for each of the three non-rank-based instruments under each of 6 recalibration families, the smallest moving R by 0.241; the two rank-based instruments are invariant across 2,400 checks to exactly zero.

4. Estimation and inference

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log. Attributes are assigned to roles by the pre-registered rules of Section 5; cases are ordered by their first timestamp; and the split is temporal. The *pipeline* axis spans encodings as well as model families: one-hot logistic regression, frequency-encoded logistic regression, cross-fitted target-encoded [25] histogram gradient boosting, and the same boosting wrapped in isotonic calibration. All four run on the case study's log; the other pairs carry two or three by a rule declared on row count. Because those levels move two things at once, Section 6.3 crosses the family with the encoding where the budget allows it, and the axis is called a pipeline rather than a learner throughout. Registers here run to 3,019 levels, above the cardinality cap of the boosting implementation's native categorical support, which is why the tree learner is target-encoded, cross-fitted inside the training half so that no training row sees its own outcome in its own encoding.

The usual interval for such an increment resamples test rows with the fitted models held fixed. It excludes training-sample variability, model-selection variability, encoder variability, split uncertainty and the temporal dependence between cases, and a design sweep is not a substitute for it: specification sensitivity and sampling uncertainty answer different questions.

Every interval in this paper is instead a *nested moving-block bootstrap.* One draw resamples the training half by moving blocks [21] of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and every quality mechanism on that resample, and evaluates on a moving-block resample of the test half. Encoders, frequency tables, target encodings, register frequency rankings and calibrators are rebuilt inside the draw. On the simulation of Section 10, where the truth is known by enumeration, the fixed-model interval covers its own estimand a median 92.1% of the time across the worlds and the nested basic interval 93.0%. Every log is also reported under rolling-origin validation — five expanding-window folds — beside the single temporal holdout, and the split is an axis of the decomposition rather than a robustness paragraph.

Refitting inside the draw breaks the percentile interval. A bootstrap resample contains about $1 - e^{-1}$ of the distinct levels of a high-cardinality categorical, so with 3,019 register levels every refit inside every draw sees a sparser register than the real training half and returns a smaller increment. The bootstrap distribution is then displaced relative to the estimate, and a percentile interval taken from a displaced distribution is a correct interval for the wrong quantity: it misses no matter how wide it is. On this corpus the displacement is small — negative in 56% of cells, by a median -0.0001 AUC

— but it bites in the regime the mechanism predicts: on the sparse world of Section 10 the percentile interval covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\,2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2}\,]$ [10], which pivots the displacement out; simultaneous bands are recentred by the same shift, which is a property of the draws and so corrects a whole family at once (Table S30, and Supplement S6 for the diagnosis).

A pivoted interval need not contain the estimate it is built from, and on this corpus 74 cells do not. Pivoting moves the interval by twice the displacement. Where the displacement exceeds the interval's own half-width, both endpoints land on the same side of $\hat{V}$ and the band excludes it. That is a property of a pivotal interval under bias rather than an error, but it is not what a reader expects of a band, so it is counted: 74 of the 3,900 whole-surface cells — 1.9% — on 3 of the 19 pairs, reaching 21.1% of the cells on the worst of them. The ratio of displacement to half-width has a median 0.09 across pairs and 0.45 on that pair.

It gets worse with the sample size, not better, and that is the uncomfortable part. The displacement is a bias — a resample holds about $1 - e^{-1}$ of the distinct levels however many rows it has — so at a fixed ratio of cardinality to sample size it does not shrink with n, while the half-width shrinks like $n^{-1/2}$. Section 10.3 locates the crossing: at a fixed ratio of 0.125 the interval straddles its own estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000. The three pairs above are the corpus's pairs nearest that crossing, and Section 6.4 reports what it costs the coverage calibration.

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* is a statement about a denominator, and the denominator has to be the set the statement ranges over. Three sets are distinguished, named, and counted from the files rather than quoted (Table S7). The **declared surface** is every scientifically admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface — 29,040 scalar cells against 29,040 declared, with 0 duplicates and 0 missing — and the audit script asserts that identity rather than the manuscript asserting it. The **inference surface** is every specification carrying bootstrap draws; it is smaller for one reason, that a draw refits the whole pipeline and its cost is the declared surface's multiplied by the draw count.

The inference surface is declared before any draw is taken, by a rule that reads only a log's row count, and it is *axis-complete*: every axis varies over at least two levels, so no axis is silently held fixed. It carries 3,900 scalar family members, a median 12.5% of each pair's computational surface, and the share of cells with a positive increment differs between the two surfaces by a median 0.067. Every quantity that does *not* need a bootstrap — the decomposition, the sign-disagreement rate, the regret comparison — is computed on the full declared surface; only the region labels are computed on the inference surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a ladder read across rungs, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} \;=\; \max_{c\in\mathcal{F}} \frac{\left|V_c^{(b)} - \bar{V}_c\right|}{\hat{\sigma}_c}, \qquad q_{1-\alpha} = \text{the } (1-\alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha}\hat{\sigma}_c$. This is the max-$t$ (Romano–Wolf) construction [29] and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger.

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells a surface-level label ranges over. Four interval objects are therefore declared, and every table says which it prints: POINTWISE, the per-cell interval, with no multiplicity control; WITHIN-INSTRUMENT, max-t over the scalar instruments at one cell, reported only so that the cost of the wider family is visible; WHOLE-SURFACE, max-t over *every* admissible scalar cell of the pair, which is the only family a surface-level claim may use; and DECISION-CURVE, max-t over every admissible net-benefit cell of the pair, declared separately because net benefit and a scalar instrument are not the same scale. The last two are the pre-registered families. The whole-surface family has a median 180 members against the five of the within-instrument family, and its median critical value is 3.33 against 2.35 and the pointwise 1.96.

Family control is **per log–target pair**: every cross-pair statement in this paper is descriptive, and none is a simultaneous inferential claim over the corpus. A replicate is dropped from a family when a resample leaves one of its cells without outcome variation, because a maximum over a moving set is not a maximum; `results/s21_incomplete.csv` names the 2 families where that happens.

Estimating the critical value without paying for it in refits. $q_{1-\alpha}$ is the upper quantile of a maximum over hundreds of correlated cells, and the empirical quantile of the B observed maxima estimates it badly, while the fitting budget cannot be raised without multiplying the cost of the whole design space by B. Standardising the draws to $Z_{b,c} = (V_c^{(b)} - \bar{V}_c)/\hat{\sigma}_c$ and multiplying them by independent standard normals gives a statistic with the empirical covariance of Z, which can be generated 20,000 times for the cost of matrix arithmetic [5]; Supplement S3 gives the construction. The fitting budget therefore fixes B and not the precision of q: the empirical quantile has a median order-statistic interval of width 2.01 at these draw counts, the multiplier quantile one of width 0.037, and the residual Monte Carlo interval is carried through, so that a cell counts as *resolved* only if it resolves at the *upper* end of it. Finally, a max-t band inherits the coverage of the interval it is built from, which Section 10.3 shows is governed by the ratio of register cardinality to training size; Section 6.4 calibrates q at each pair's own ratio.

And the two estimates of the same critical value disagree, in the direction that matters. The precision argument above is an argument about variance, and the two estimators differ by more than variance. On 15 of the 19 whole-surface families, and on *every* one of the 19 decision-curve families, the multiplier value lies below the entire order-statistic interval of the empirical quantile: the median ratio q_{emp}/q is 1.29 on the surface families and 2.09 on the decision-curve ones, whose medians are 3.87 against 8.10. The direction is anti-conservative — a band built on the smaller value resolves cells the observed maxima do not — and the mechanism is visible in the draws. A Gaussian multiplier reproduces the empirical *covariance* of the standardised draws and therefore only the excursions a Gaussian process makes. These draws are heavy-tailed — Section 10.4 measures how, and finds that the word covers two different mechanisms on the two families — and the asymptotics the construction rests on are asked to hold at 40 to 150 draws over families of several hundred cells.

We therefore report what each estimator resolves rather than asserting one of them. Over the whole-surface families the multiplier band resolves 1,150 cells and the empirical band 995; over the decision-curve families, 3,066 against 613, which is the larger exposure and belongs to the one family the coverage calibration of Section 6.4 does not touch. Section 6.4 carries the surface counts under both and Section 8.3 the decision-curve ones.

Which of them is right, measured against a known answer. A disagreement between two estimators of one quantity says that one of them is wrong and cannot say which, so Section 10.4 measures the family-wise coverage each one attains over synthetic families whose size, draw count, cross-cell correlation and tail behaviour are matched to this corpus, and whose truth is known by construction. Three results govern how the bands in this paper should be read, and none of them is comfortable.

The multiplier band does not attain its nominal level anywhere in this corpus's configuration, and the shortfall is not a tail effect. Under *Gaussian* draws — the case the multiplier is derived for — its family-wise coverage has a median 87.3% over the matched cells and a minimum 74.8% against a nominal 95%; under draws matched to this corpus's tails, 84.6% and 76.1%. The construction is asymptotic and these draw counts are not asymptotic.

The safeguard the resolved flag uses buys almost nothing. Resolving a cell only at the upper end of the critical value's Monte Carlo interval moves coverage from 84.6% to 84.8%, because that interval is narrow by design — it is the precision the multiplier was adopted for. It controls the Monte Carlo error in q and not the error in what q estimates.

The best of the five candidates is the empirical quantile taken at the upper end of its own order-statistic interval, and it is still short. It reaches a median 92.2% and a minimum 86.5% on the matched cells, at 1.52 times the width. A Rademacher wild multiplier, which reuses the observed draws with random signs instead of replacing them by Gaussians and which we expected to repair exactly this, does worse than the Gaussian one: 80.7%. Section 10.4 reports all five.

The binding constraint is the number of draws, not the choice among estimators. At the modal surface family the multiplier band covers 82.0% at 33 draws, 90.9% at 80, 94.4% at 150 and 95.8% at 400; the per-cell interval it is built from moves 90.3% to 94.6% over the same range. This corpus runs at 40 to 150 draws as declared and 33 to 150 effectively, and it is the effective count a band is built from. Every band, region label and ρ in this paper should be read as the anti-conservative statement that makes it, and Section 11 states what follows.

4.4. Planned contrasts, and a p-value the design can produce

Five contrasts on the primary log are *planned for this analysis round*: that the register adds to the intake baseline (PC1); that it still adds once the free field is admitted (PC2); that it adds at the later decision time (PC3); that the absorption D is positive (PC4); and that a half-empty register is worth less than a full one (PC5). Each is stated with its complete estimand — log, cohort, target, learner, encoding, split, decision time, register quality, baseline and instrument — in Table S8. They are not *confirmatory*: they were fixed before this analysis round, not before the data were seen. The p-values are one-sided bootstrap p-values with a Holm step-down correction [15] at $\alpha = 0.05$, computed with the plus-one estimator $\hat{p} = (k+1)/(B+1)$, where k counts draws on the null side; a p-value of zero is a number no finite bootstrap can produce. At 2,000 draws the smallest attainable Holm-adjusted value over five tests is 0.00250, and 5 of the five survive.

A one-sided p-value and a two-sided basic interval disagree on PC3, and the interval decides. PC3's one-sided p is 0.00100 (Holm-adjusted 0.00250) and its two-sided basic interval, $[-0.00059, +0.01210]$, contains zero. Both come from the same 2,000 draws. The p-value says where those draws lie — their central ninety-five per cent runs $+0.00327$ to $+0.01596$, entirely above zero — while the interval says where the *estimand* lies, obtained by pivoting that distribution about the estimate; the draws sit 0.00193 AUC *above* the estimate, the upward shift Section 4.1 diagnoses, so pivoting moves the interval down by twice that shift and its lower limit crosses zero even though almost no draw does. **A contrast is called resolved only when the interval says so**, and PC3 is not resolved on any of the four cohort-and-target combinations Section 7.1 reports. Every other comparison here is *exploratory*, labelled as such where it appears, and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

4.5. The sensitivity decomposition

Because the surface is a complete factorial over its axes — the audit of Section 4.2 establishes that it is, cell for cell — the variance of V_s across specifications admits an exact and *orthogonal* functional-ANOVA decomposition [33, 30] under a product measure $\mathbb{P} = \bigotimes_i w_i$ over the axes. The closest precedent is Hutter et al. [16], who decompose an algorithm's performance over its *configuration* space by the same device and read the first-order and interaction terms as hyperparameter importance. What differs here is the object and what is done with it: the axes are analysis decisions rather than hyperparameters, the components are computed exactly over a complete factorial rather than estimated from a regression-tree surrogate over a sampled design, and the decomposition is one of four reporting objects rather than the conclusion. Every subset u of axes has a component g_u, given by a Möbius sum over the conditional means and computed exactly rather than estimated (Supplement S3); the components are mutually orthogonal, so their variances D_u satisfy $\sum_{u \neq \emptyset} D_u = \mathrm{Var}(V)$ exactly. Writing $S_u = D_u / \mathrm{Var}(V)$, the first-order and total indices are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u \ni i} S_u$.

The per-axis difference $S_{T_i} - S_i$ is the variance in which axis i is *involved* beyond its own main effect. Those quantities overlap across axes, because a two-way component appears in the total effect of both of its axes, so their largest member is not the share of variance carried by interactions. That share is

$$S_{\text{int}} \;=\; 1 - \sum_i S_i, \tag{3}$$

which this paper computes together with every disjoint component, so the identity $\sum_u S_u = 1$ is asserted numerically — it holds on all 304 decompositions — rather than assumed. The other quantity is retained as the *largest per-axis interaction involvement*; the two differ substantially, at 43.7% against 56.0% at the median pair.

The measure the decomposition is taken under. A variance decomposition is taken with respect to a distribution over specifications, and *uniform over the cells I happened to enumerate* is a choice rather than a neutral default: it moves if an analyst adds a pipeline, writes five folds as five levels, or refines a threshold grid. Axis weights are therefore separated from within-axis level weights and every quantity is reported under declared measures (Table S19): EQUAL-LEVEL, every level of every axis equally likely; REFERENCE, half the mass on the reference level of each axis; CONCENTRATED, four fifths on it; and an ENVELOPE over Dirichlet draws of the level weights. Two invariance properties are executed rather than asserted: duplicating a level with its weight split leaves every component unchanged, to 5.6×10^{-17}, and refining the operating-point grid under the same underlying continuous measure moves the threshold index by 0.004.

Indices are estimated quantities: the whole decomposition is recomputed inside every draw the inference surface supplies, corpus-level medians are bootstrapped with the pair as the resampling unit, and no axis is called dominant where the intervals overlap. They are computed within each instrument, in its own units, which is the primary analysis, and on the *headroom* scale $V_s/(1 - m(B))$, which is comparable across instruments and lets the metric enter as an axis. The second is a sensitivity analysis rather than a uniquely meaningful decomposition, because of the following.

Proposition 3 (The decomposition inherits the metric's scale)**.** *Let ϕ be strictly increasing with $\phi(0) = 0$. The first-order sensitivity indices of V and of $\phi(V)$ over the same design need not agree, and there exist designs and ϕ under which one axis has the larger index on one scale and a different axis on the other.*

The 3×3 construction is in Supplement S2. This is the same phenomenon as Proposition 1 one level up: just as the reduction is not identified without the metric, "which choice matters most" is not identified without the scale on which the increment is expressed, so the manuscript never says an axis dominates without naming it.

4.6. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions)**.** *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws, a median 12.5% of that pair's admissible scalar cells. Under the simultaneous band of Section 4.3, a cell of $\mathcal{F}$ is* beneficial *if its lower band is above zero,* harmful *if its upper band is below zero, and* unresolved *otherwise. A surface is* uniformly beneficial *if every cell of $\mathcal{F}$ is beneficial;* conditionally beneficial *if some are and none is harmful;* uniformly harmful *and* conditionally harmful *in mirror image;* sign-changing *if there is at least one cell of each kind; and* unresolved *if none is resolved. The* robustness index *is*

$$\rho \;=\; \frac{\#\{\textit{beneficial}\} - \#\{\textit{harmful}\}}{|\mathcal{F}|} \;\in\; [-1, 1].$$

The label is relative to $\mathcal{F}$ and not to the whole admissible set, and the direction of that is not neutral. A label computed on a sixth of a

pair's cells is systematically cleaner than one computed on all of them, and *uniformly beneficial* is the label most helped by it: a claim about every cell is easier to sustain over fewer cells. The manuscript therefore never says a surface is uniformly beneficial without this restriction being in force, and Section 6.4 reports that the label is not reached on any pair in any case.
The two harmful labels are not decoration: Table 5 carries 1 conditionally harmful and 0 uniformly harmful surfaces out of 19. $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and the empirical question this paper answers is how often it obtains. The region is defined over the *scalar* family; folding the decision-curve cells into it would let a single extreme threshold decide the label of a whole surface, so the decision curve gets its own region (Section 8). The intercept-only rung is excluded from every admissible set, because a reduction measured against a model with no features is inflated by construction; the master surface file *does* carry that rung, which is why the two counts in Table S7 differ.

4.7. Specification regret

A reader of a one-number claim decides whether to adopt the field, and the rule the number supports is *adopt iff* $V_s > 0$. If the specification the reader will actually run is drawn from the same admissible set under the declared measure, two quantities follow. The *misreport rate* $M(s) = \Pr_{s'}[\operatorname{sign} V_{s'} \neq \operatorname{sign} V_s]$ is the probability that the reader's own cell disagrees in sign with the reported one. The *regret*

$$G(s) \;=\; \mathbb{E}_{s'}\Big[\max(0, -V_{s'})\Big]\, \mathbf{1}\{V_s > 0\} \;+\; \mathbb{E}_{s'}\Big[\max(0, V_{s'})\Big]\, \mathbf{1}\{V_s \leq 0\}$$

is the performance destroyed by adopting when the reader's cell is harmful plus the performance forgone by declining when it is beneficial.

$G(s)$ carries the units of the instrument its increments were measured in, so an average across five scalar instruments has no units at all. Three designs are therefore reported, each constructed in Supplement S3: **A, metric-specific**, inside one instrument at a time, which is what the headline uses; **B, common operational utility**, on calibrated net benefit (Section 8.4); and **C, partial identification**, over five declared monotone utility maps fixing zero. The misreport rate has a second difficulty, which Section 6.5 answers rather than notes: a cell whose sign the data do not resolve contributes a coin flip to it. The rate is therefore reported over all cells, over the cells the calibrated band resolves, and weighted by $|V|$, and the article's headline is the second.

Saying that a surface report asserts a sign only where the region is uniform, and therefore misreports nothing, is a definitional win and not a measured one, so the paper compares explicit *decision rules* under the loss above.

Lemma 1 (The decision-optimal collapse)**.** *Let the reader's specification be drawn from the admissible set $\mathcal{S}$ under a weighting w, and let a rule choose one action for all readers. Writing $A = \mathbb{E}_w[\max(0, -V_s)]$ for the harm mass and $B = \mathbb{E}_w[\max(0, V_s)]$ for the benefit mass, expected loss is A under* ADOPT *and B under* DECLINE*, and $B - A = \mathbb{E}_w[V_s]$. Hence expected loss is minimised by adopting exactly when $\mathbb{E}_w[V_s] > 0$: the decision-optimal summary of a specification surface is the* weighted mean increment*, not the sign at any one cell and not the count of positive cells.*

Its assumptions are strong — one action for every reader, a loss linear in a common utility, and a known reader distribution w — and what is scientifically difficult is specifying the utility and the weighting, not proving the identity. The paper therefore reports the identity as a check — the mean rule's in-sample excess regret is exactly zero across every pair and instrument, as it must be — and measures the rules *out of sample* instead (Section 6.8). Two things follow: the conservative rule is wrong, not merely cautious, whenever the harm mass is smaller than the benefit mass it forgoes; and a one-number report is optimal exactly when the sign of its cell agrees with the sign of the mean, which is a coincidence rather than a property of the cell.

5. The registered benchmark design

5.1. What was fixed, and when

The rules that turn an arbitrary event log into an instance of this problem were committed before any result on a new log existed, in `PROTOCOL.md` for the corpus and `AUDIT-PROTOCOL-2.md` for the practice pilot. The version history shows the registering commit adding one file and no result file; `git log --stat` is the check, and `REPRODUCE.md` names the commit. Six amendments were declared afterwards, each with its date, its cause and the result it changed, and all six are in Supplement S1 rather than folded silently into the rules.

This is not a timestamped third-party registration, and the difference matters. A registration on a public registry is attested by a party with no stake in the result; what is offered here is a file in a repository whose history the authors control, so a reader who does not trust that history has nothing else to check. Any claim in this paper that a rule was fixed in advance is therefore a claim about the repository and no stronger than the repository. We say so plainly instead of letting the word *registered* carry a weight it has not earned: what the design buys is that the rules are legible, dated and amendable only on the record, not that an outside party watched them being written.

5.2. Roles

A log is admitted if the registered rules can find three things in it without reading an outcome.

The register f. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field g. A resource-like attribute, present on at least 50% of cases, whose value varies within a case, so that it can carry information about routing.

The intake block B_0. Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f.

Coarsenings of f — its type, its subtype, the service it belongs to — are given their own role rather than being swept into B_0, because which *layer* of a register pays is a separate question, answered in Supplement S4.

5.3. Targets, splits and exclusions

Two targets are defined for every log: *handover*, whether more than one resource group touches the case, and *duration*, whether the case runs longer than the training half's median. Cases are ordered by their first timestamp and split 70/30; the rolling-origin scheme adds five expanding-window folds. Fields that are determined only at closure are excluded by name, which is the standard leakage discipline [19, 18] and, for event logs specifically, the discipline of Weytjens and De Weerdt [44]. A log–target pair is excluded only for a registered reason — too few cases, no variation in the test half, a prevalence outside $[0.05, 0.95]$, or the absence of one of the three roles —

and every exclusion is reported with its code in Table S5, which accounts for every downloaded log that is not among the admitted pairs — including the 1 that were downloaded as a held-out set for a separate test and never offered to these rules at all.

Every admitted pair is analysed and reported. Running the full surface only on the logs whose *reduction* is resolvable would condition the analysis on the outcome. The surface is computed on all 19 admitted pairs through one code path, and the pairs on which the register is worth nothing are in the same tables as the rest.

5.4. What the corpus does and does not have in common

The corpus is 13 public logs spanning 6 domains, and it includes the further ITSM logs the literature uses [35, 2]. It is heterogeneous and the paper does not claim otherwise. The registers are a configuration item, a caller, a product, a vendor, a diagnosis and a legal article; the free fields are an assignment group, an assignee, a resource and a role; the targets mean different things in a bank's incident queue, a hospital's admissions and a municipality's permits.

Table S6 makes that explicit rather than leaving it to a sentence. For every pair it names the domain, the field playing the register role, why that field behaves as a maintained and reused entity, the free field, the interpretation of the target, and the specific threat to construct validity that the pair carries. A diagnosis code is admitted as a register by a cardinality-and-reuse rule, and it is not a configuration management database; a legal article is reused across cases but nobody maintains a discovery process for it. Those are threats, they are named per pair, and the paper's conclusions are stated at a level that survives them.

The pair carrying the most cells sits closest to the admission edge. BPIC14/handover has the highest prevalence of any admitted pair, 0.927, against the window's upper bound of 0.95; the nearest pair the headroom rule turns away, BPIC15_5/handover, is at 0.959, and 5 pairs in all are excluded above the bound. That pair is also the largest single contributor to the corpus, carrying 4,800 of the 29,040 admissible scalar cells. A rule with a sharp edge decides some case narrowly, and this is that case: the corpus's biggest component is admitted on a margin narrower than the gap to the pairs excluded beside it. Table S10 reports every headline statistic on the

ITSM-like subset as well, which is the population that does not turn on this decision.

This is therefore a *benchmark of specification sensitivity across heterogeneous prediction problems*, not a replication of one finding on several organisations, and two disciplines follow. No domain-specific effect magnitude is averaged across domains as though it estimated one universal register value: the corpus-level statistics are all about *sensitivity*, which is a property each pair has separately and which can be summarised across pairs without assuming they measure one thing. And the 8 ITSM-like pairs, the closest family to the case study, are analysed as a group as well, with *every* headline corpus statistic reported on both populations in Section 6.6.

6. Multi-log results

Every point estimate in this section comes from one master surface, `results/s01_surface.csv`, computed by one pass of one pipeline over 19 log–target pairs, 4 learners, 6 splits, 10 register-quality conditions, the nested baseline ladder and 36 instruments including the 31-point decision-curve grid. Its 273,024 rows *include* the intercept-only rung, so that the surface is complete; no admissible set contains it, and the 29,040 admissible scalar cells every headline is computed on exclude it. Every interval is the basic construction of Section 4.1 and every band the whole-surface simultaneous construction of Section 4.3, and each table names which it prints. The denominator audit is in Table S7: declared equals computed on all 19 pairs, with 0 duplicates and 0 missing, and the inference surface (3,900 scalar members) is a subset of it cell for cell, which `verify_numbers` asserts.

6.1. The master table

Table 3 is the frozen master table: one row per log–target pair, the increment at the reference specification with its simultaneous band, the region label, the robustness index and the sign-disagreement rate of a conventional one-number report. The reference specification is declared and is the same everywhere — one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field as the baseline, ROC AUC — and its region and ρ columns are the coverage-calibrated ones of Section 6.4 while its sign-disagreement column is the all-cells rate of Table 6 under the equal-level measure.

log	target	V at reference	pointwise	simultaneous	region	rho	misreport rate
BPIC13_closed	handover	-0.009	[-0.041, +0.018]	[-0.068, +0.046]	unresolved	0.000	0.284
BPIC13_incidents	duration	0.028	[+0.013, +0.060]	[-0.003, +0.074]	conditionally beneficial	0.017	0.393
BPIC13_incidents	handover	0.037	[+0.030, +0.050]	[+0.021, +0.059]	conditionally beneficial	0.422	0.111
BPIC14	duration	0.096	[+0.089, +0.109]	[+0.081, +0.119]	conditionally beneficial	0.692	0.106
BPIC14	handover	0.167	[+0.143, +0.205]	[+0.115, +0.228]	sign-changing	0.552	0.244
BPIC15_1	duration	0.001	[-0.029, +0.046]	[-0.073, +0.071]	conditionally beneficial	0.011	0.423
BPIC15_1	handover	0.007	[-0.007, +0.021]	[-0.017, +0.032]	unresolved	0.000	0.242
BPIC15_3	duration	0.035	[-0.000, +0.070]	[-0.025, +0.094]	conditionally beneficial	0.056	0.191
BPIC15_3	handover	-0.003	[-0.047, +0.058]	[-0.088, +0.088]	unresolved	0.000	0.616
BPIC15_4	duration	0.035	[+0.016, +0.084]	[-0.008, +0.101]	unresolved	0.000	0.623
BPIC15_4	handover	0.001	[-0.026, +0.056]	[-0.059, +0.076]	unresolved	0.000	0.618
BPIC15_5	duration	0.032	[-0.026, +0.100]	[-0.075, +0.143]	unresolved	0.000	0.328
BPIC17	duration	0.002	[-0.003, +0.010]	[-0.008, +0.018]	conditionally beneficial	0.139	0.353
BPIC19	duration	0.024	[+0.011, +0.043]	[-0.001, +0.052]	sign-changing	-0.083	0.360
Helpdesk	duration	-0.020	[-0.039, -0.003]	[-0.054, +0.007]	conditionally harmful	-0.011	0.257
RoadFines	duration	0.031	[+0.025, +0.037]	[+0.020, +0.042]	sign-changing	0.225	0.173
Sepsis	duration	0.153	[+0.114, +0.210]	[+0.076, +0.237]	conditionally beneficial	0.283	0.056
UCI498	duration	-0.007	[-0.011, +0.006]	[-0.016, +0.015]	conditionally beneficial	0.094	0.493
UCI498	handover	-0.003	[-0.007, -0.001]	[-0.009, +0.002]	conditionally beneficial	0.044	0.481

Table 3: The master table. One row per admitted log–target pair. 'pointwise' is the 95% interval at the reference cell, basic (pivotal) construction; 'simultaneous' is that same cell's WHOLE-SURFACE simultaneous band, at the conservative end of the critical value's Monte Carlo interval. They are different objects, and the region label uses only the second. Then the resolution region and the robustness index ρ, both under the COVERAGE-CALIBRATED critical value of Section 6.4 — the nominal ones are in Table 5 — and the share of admissible specifications whose sign disagrees with a conventional one-number report, under the equal-level measure, which is the 'all-cells rate' column of Table 6. The reference cell's sign and the region need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative.

6.2. What the choices are worth

Table 4 and Figure 2 give the decomposition, computed within each instrument in that instrument's own units, under the equal-level measure. The table's first four columns are, per pair, the median over the five instruments of each axis's first-order index; its last column is the largest per-axis interaction *involvement*. Three aggregations of this table are quoted below and they are not interchangeable, so each is named where it appears: the median over pairs of a column of this table; the 28.7% figure, which is the median over pairs of the median over instruments of the *per-instrument* largest first-order index, and which is therefore not obtainable by taking a maximum across the four printed columns; and the corpus medians of the higher-order share. The per-instrument quantities the second is built from are in Table S19.

No single axis dominates, and the largest is the one an evaluation paper is most likely to fix without comment. Across the corpus the median first-order index is 18.9% for the split, 5.5% for the baseline rung,

log	target	learner	quality	rung	split	higher-order total	largest involvement
BPIC13_closed	handover	0.110	0.029	0.075	0.203	0.560	0.437
BPIC13_incidents	duration	0.036	0.029	0.029	0.384	0.409	0.372
BPIC13_incidents	handover	0.024	0.099	0.169	0.172	0.308	0.251
BPIC14	duration	0.242	0.151	0.359	0.013	0.233	0.180
BPIC14	handover	0.314	0.127	0.307	0.010	0.233	0.183
BPIC15_1	duration	0.041	0.039	0.055	0.263	0.614	0.548
BPIC15_1	handover	0.018	0.036	0.075	0.079	0.575	0.476
BPIC15_3	duration	0.003	0.081	0.025	0.287	0.607	0.490
BPIC15_3	handover	0.002	0.004	0.128	0.189	0.662	0.547
BPIC15_4	duration	0.060	0.025	0.011	0.111	0.732	0.655
BPIC15_4	handover	0.022	0.031	0.035	0.050	0.793	0.643
BPIC15_5	duration	0.078	0.042	0.057	0.036	0.749	0.664
BPIC17	duration	0.000	0.062	0.001	0.346	0.590	0.494
BPIC19	duration	0.014	0.011	0.003	0.491	0.474	0.418
Helpdesk	duration	0.262	0.086	0.008	0.222	0.339	0.314
RoadFines	duration	0.009	0.289	0.032	0.192	0.449	0.424
Sepsis	duration	0.021	0.145	0.008	0.134	0.608	0.574
UCI498	duration	0.048	0.030	0.056	0.399	0.433	0.391
UCI498	handover	0.079	0.071	0.287	0.136	0.390	0.285

Table 4: The corrected sensitivity decomposition, within instrument, under the equal-level measure. The four first-order indices, then the TOTAL higher-order share $1 - \sum_i S_i$ of Equation (3), then the largest per-axis interaction involvement $\max_i(S_{T_i} - S_i)$ — which is the quantity an earlier version reported as the interaction share and which is not it, because those differences overlap across axes.

4.2% for the register-quality condition and 3.6% for the pipeline. The split leads the other three by a wide margin and still explains well under a quarter of the variance. At the median pair the largest of the four, whichever it is, is 28.7%.

Most of the variance belongs to no single axis. The total higher-order share $1-\sum_i S_i$ has median 56.0% (95% interval 40.9% to 60.8% over pairs), ranging from 23.3% to 79.3%; the largest per-axis *involvement* $\max_i(S_{T_i} - S_i)$ is a different quantity and is 43.7% at the median pair.

The measure is not neutral, and one conclusion does not survive it. Under the reference-weighted measure the median higher-order share is 49.6% and under the concentrated measure 31.7% (Table S19). That second number is smaller than the largest first-order index under the same measure, 39.2%, and on 12 of 19 pairs some single axis carries more variance than everything higher-order combined — against 6 pairs under the equal-level measure. The statement *the interactions carry more than any main effect* is therefore true under the two measures that let a reader roam and false under the one that keeps them near the reference cell. Both invariance tests pass,

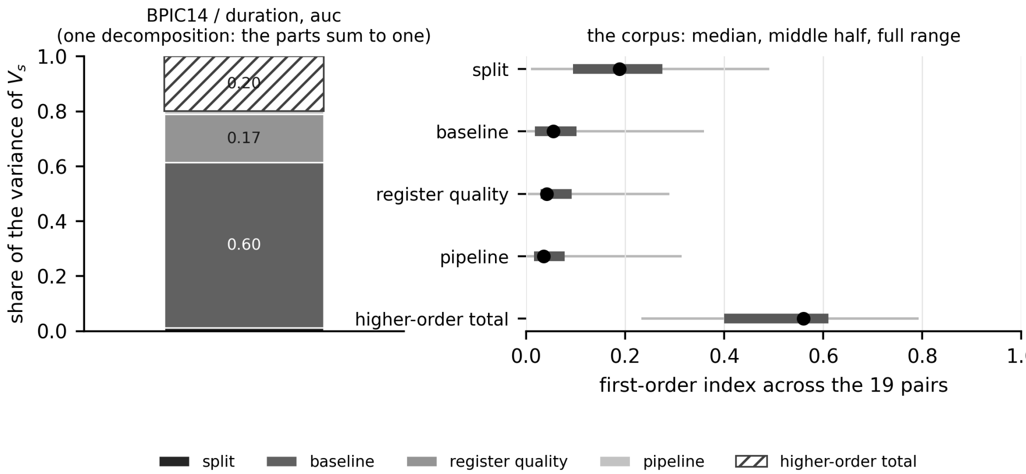

Figure 2: The decomposition, within instrument, under the equal-level measure. *Left*: one pair at one instrument, so the components are a decomposition and the parts sum to one exactly — the largest pair in the corpus, at the reference instrument, which is the cell every other table reports at. The hatched segment is the total higher-order share $1 - \sum_i S_i$ of (3). *Right*: each axis's first-order index across the corpus, as the median over pairs with the range the middle half occupies and the full range. Nothing on the right is stacked, because a median over pairs and a median over instruments do not add and a stacked bar of them would invite the addition. The bootstrapped indices and their intervals are in Table 4.

to 5.6×10^{-17} and 0.004.

The target is an axis too, and it is a small one within a pair. Every index above is computed *within* a log–target pair. On the 6 logs carrying both registered targets the target axis can be added, on the headroom scale, and it carries a median 2.1%. The two targets are different questions, agreeing on a median 51% of cases, so the reading is not that the target hardly matters — on the case study it decides the sign (Section 7.1) — but that once it is fixed, the remaining choices move the answer by more than switching it does.

Which axis leads is a property of the pair. The largest single first-order index anywhere on the corpus is 77.0%, carried by the baseline rung on UCI498 / duration; 4 different axes lead somewhere, and the most frequent leader is the split, on 51% of pair–instrument combinations. An analyst can guess the leader and be right about half the time.

The baseline axis moves the answer by more than the answer. Holding the pipeline, the split, the register and the metric at the reference and varying only the baseline rung, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself at the reference cell on 15 of the 19 pairs. The index and the spread answer different questions, what share of the *variance* an axis explains while every other axis is also varying against how far that one axis moves the answer on its own, and both are reported because either alone misleads.

6.3. The pipeline axis is two axes

Table 4's "learner" column is not a learner axis. Its levels are pipelines: a model family together with an encoding and, in one case, a calibration, and on 15 of the 19 pairs only two of them run. On the case study's log the budget allows the two to be crossed — both families against all three encodings on both targets, 12 complete factorial pipelines — and the crossed decomposition separates them (Supplement S10, Table S4). **The encoding is the larger of the two**, at 10.1% of the variance against the family's 5.2%, and their interaction, 9.7%, is as large as either main effect. The fused index is therefore not the sum of two indices. The crossing has not been re-run on the other 15 pairs, and Section 11 carries that as a limitation.

6.4. Resolution regions, and the coverage they have

Table 5 and Figure 3 carry the region labels and the robustness index for every pair, computed on the inference surface under the whole-surface simultaneous band at the conservative end of the critical value's Monte Carlo interval. That family has a median 180 members per pair and a median critical value of 3.33, against 2.35 for a within-instrument family and 1.96 pointwise; under the narrower family the same draws give a different region label on 9 of the 19 pairs, and the whole-surface family resolves 1,150 cells against the narrow family's 1,510.

The nominal critical value is not enough, and the correction is measured rather than assumed. Section 10.3 shows that coverage is governed by K/n and that this corpus lives where coverage is below nominal, the median pair sitting at 0.106, so a region label computed from a nominal band overstates how much of a surface is resolved. The (n, K) plane is dense enough to calibrate against, and Supplement S9 derives the inflation factor $c(K/n)$ from it — a monotone regression through the measured factors, whose log-linear summary has slope 0.102 (standard error 0.005). The factors the corpus draws range from 1.18 to 1.77; at the median pair the critical value goes from 3.33 to 4.54; the corpus resolves 896 cells against 1,150, a loss of 22.1%, and 4 of the 19 region labels change. **Every region label and**

log	target	K / n	q upper	c	q calibrated	cells (nominal)	rho	cells (cal)	rho (cal)	region
BPIC13_closed	handover	0.324	3.359	1.767	5.935	0 / 1 / 179	-0.006	0 / 0 / 180	0.000	unresolved
BPIC13_incidents	duration	0.133	3.374	1.456	4.914	21 / 0 / 159	0.117	3 / 0 / 177	0.017	conditionally beneficial
BPIC13_incidents	handover	0.133	3.348	1.456	4.876	97 / 0 / 83	0.539	76 / 0 / 104	0.422	conditionally beneficial
BPIC14	duration	0.093	3.509	1.294	4.540	337 / 0 / 143	0.702	332 / 0 / 148	0.692	conditionally beneficial
BPIC14	handover	0.093	3.589	1.294	4.643	279 / 2 / 199	0.577	267 / 2 / 211	0.552	sign-changing
BPIC15_1	duration	0.106	3.357	1.343	4.509	6 / 0 / 174	0.033	2 / 0 / 178	0.011	conditionally beneficial
BPIC15_1	handover	0.106	3.382	1.343	4.542	2 / 0 / 178	0.011	0 / 0 / 180	0.000	unresolved
BPIC15_3	duration	0.095	3.353	1.300	4.358	40 / 0 / 140	0.222	10 / 0 / 170	0.056	conditionally beneficial
BPIC15_3	handover	0.095	3.339	1.300	4.341	2 / 0 / 178	0.011	0 / 0 / 180	0.000	unresolved
BPIC15_4	duration	0.072	3.400	1.243	4.227	0 / 0 / 180	0.000	0 / 0 / 180	0.000	unresolved
BPIC15_4	handover	0.072	3.447	1.243	4.286	0 / 0 / 180	0.000	0 / 0 / 180	0.000	unresolved
BPIC15_5	duration	0.132	3.412	1.455	4.964	4 / 3 / 173	0.006	0 / 0 / 180	0.000	unresolved
BPIC17	duration	0.032	3.271	1.177	3.849	35 / 0 / 145	0.194	25 / 0 / 155	0.139	conditionally beneficial
BPIC19	duration	0.011	3.336	1.177	3.926	14 / 28 / 78	-0.117	14 / 24 / 82	-0.083	sign-changing
Helpdesk	duration	0.123	3.333	1.431	4.769	0 / 8 / 172	-0.044	0 / 2 / 178	-0.011	conditionally harmful
RoadFines	duration	0.001	3.158	1.177	3.717	47 / 20 / 53	0.225	45 / 18 / 57	0.225	sign-changing
Sepsis	duration	0.200	3.338	1.562	5.215	106 / 0 / 74	0.589	51 / 0 / 129	0.283	conditionally beneficial
UCI498	duration	0.301	3.276	1.756	5.751	58 / 0 / 122	0.322	17 / 0 / 163	0.094	conditionally beneficial
UCI498	handover	0.301	3.349	1.756	5.880	40 / 0 / 140	0.222	8 / 0 / 172	0.044	conditionally beneficial

Table 5: Resolution regions under the nominal critical value and under one calibrated for the register's cardinality. c is the inflation factor the (n, K) plane of Section 10.3 gives at that pair's own K/n; the calibrated critical value is c times the nominal one. 'q upper' is the UPPER end of the critical value's Monte Carlo interval, which is the value the bands use by the rule of Section 4.3 that a cell resolves only at the conservative end; the point estimate whose corpus median the text quotes is smaller. The cell columns are beneficial / harmful / unresolved, and the admissible count is the same under both critical values. The calibrated columns are the ones the article's counts use; the nominal ones are printed beside them so that the size of the correction is visible.

every ρ quoted in this paper is the calibrated one; Table 5 prints the nominal columns beside them.

The factor's reach in n, and the two things widening the plane found. The plane is fitted at training sizes 500 to 180,000 and this corpus runs 735 to 176,213, so all 19 of the 19 pairs lie inside the range the factor is identified on. That is the arrangement a calibration needs, and it is not the end of the matter.

K/n is not sufficient, and a plane wide enough in n can now show it. Adding $\log n$ to the fit gives a coefficient of 0.067 at $t = 8.3$: at a fixed ratio the widening a pair needs grows with its sample size. The design that isolates it is the ladder, three ratios held fixed while n moves over two orders of magnitude, and along it the measured factor moves by 0.22 — 2.8 combined Monte Carlo standard errors — while coverage itself moves by 21.7%. The applied curve is a function of the ratio alone, so it carries that dependence as a residual: it sits below the measured factor's own lower confidence limit on 2 of the plane's cells, by at most 0.18, and both are large-n cells. A two-covariate fit in $\log(K/n)$ and $\log n$ does not repair it — it is worse than the monotone curve on every cell-weighted criterion (Supplement S9) — so the

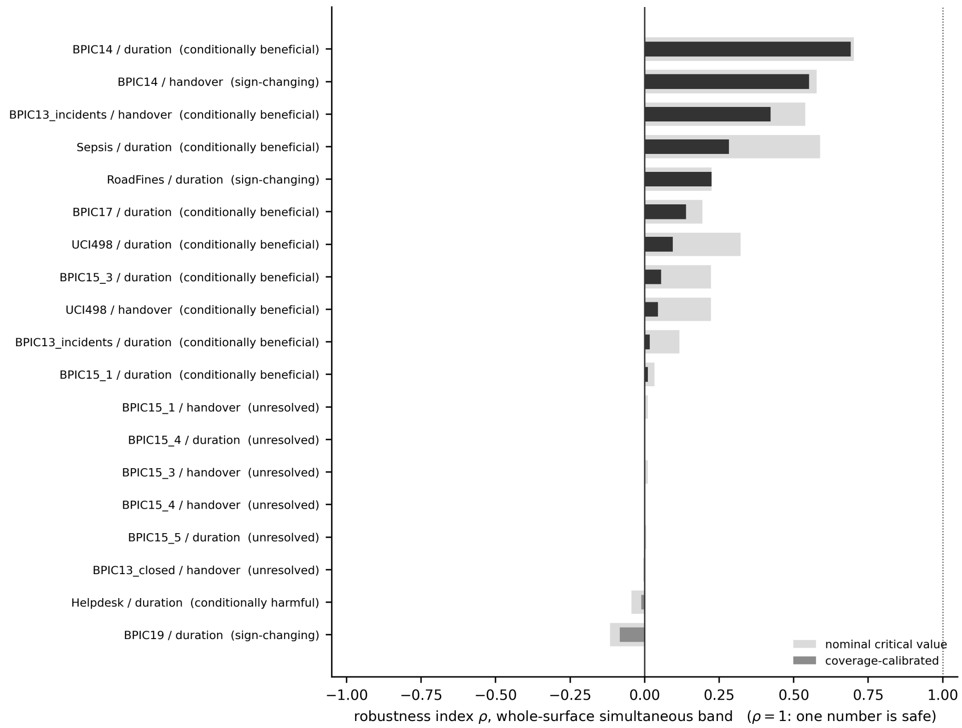

Figure 3: The robustness index ρ per log–target pair, with the region label, under the coverage-calibrated critical value. $\rho = 1$, the dotted line, is the only state in which a single positive number is a safe summary.

residual is reported rather than fitted away.

And on 4 of the 33 cells the factor does not exist at all. The inflation factor is a widening *about the point estimate*, and Section 4.1 shows that past a crossing in n the basic interval stops containing the point estimate. Where it does, there is no widening about $\hat{V}$ that attains nominal coverage and the quantity this calibration estimates is undefined. All 4 such cells are at 50,000 training rows or more. The admission rule — a cell enters the plane when the interval straddles its estimate in at least 0.90 of replicates — decides nothing: the rejected cells reach 0.078 and the admitted ones start at 0.976, so any threshold between those two gives this plane. Section 11 carries what that leaves the corpus's largest pairs.

Under the calibrated band the region counts are: conditionally beneficial, 9; sign-changing, 3; conditionally harmful, 1; unresolved, 6; and the median robustness index is 0.017.

Of the 13 pairs that resolve anything, 9 have no harmful cell and 3 resolve cells of both signs. That is the informative reading of the region column, and it is the one to carry away: on two thirds of the pairs where the data determine any sign at all, every sign they determine points the same way; on the remaining third they do not.

No surface in this corpus is uniformly beneficial, and that statement is weaker than it looks. It holds under the calibrated band, the nominal one and the empirical one — but *uniformly beneficial* requires every admissible cell's lower band to be above zero, and a cell whose point estimate is negative can never qualify at any critical value. Every one of the 19 pairs contains at least one such cell, so the label is unreachable here before any band, bootstrap or calibration is applied, and the count of zero is not an inferential finding. The statement that carries content needs none of the apparatus: on every pair at least one admissible specification has a negative increment, which is visible from the all-cells column of Table 6. What the apparatus earns is ρ and the resolved counts, not this. For the same reason $\rho = 1$ is not an attainable state for any pair in this design, and Section 4.6's remark that it is the only safe state for a single number should be read as a definition rather than as a hypothesis this corpus tests.

The counts are reported at four settings, and the ordering is stable across all of them. At $c = 1$ — the nominal band, no calibration at all — the corpus resolves 1,150 cells at a median ρ of 0.117; under the calibrated factors, 896 cells at 0.017; pushing every factor to the upper end of its own Monte Carlo interval, 833 at 0.011; and under the *empirical* critical value of Section 4.3 rather than the multiplier one, 995 against the multiplier's 1,150. That last comparison is not a calibration setting but a different estimator of the same quantity, and it moves the resolution by more than the calibration does. What moves across all four is how much resolution the corpus has; what does not move is that no pair reaches $\rho = 1$. The transfer from a pointwise interval in a simulation to a simultaneous band on real logs, and the choice between the two critical-value estimators, are both assumptions, and Section 11 carries them as limitations.

6.5. What one number costs

A conventional report — one pipeline, the clean register, the single holdout, ROC AUC — disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. Two things are wrong

log	target	cells	all-cells rate	resolved / of which disagree	resolved rate	magnitude rate
BPIC13_closed	handover	1080	0.284	0 / 0	–	0.111
BPIC13_incidents	duration	1080	0.393	3 / 0	0.000	0.812
BPIC13_incidents	handover	1080	0.111	76 / 0	0.000	0.482
BPIC14	duration	4800	0.106	332 / 0	0.000	0.018
BPIC14	handover	4800	0.244	269 / 2	0.007	0.025
BPIC15_1	duration	1080	0.423	2 / 0	0.000	0.502
BPIC15_1	handover	1080	0.242	0 / 0	–	0.529
BPIC15_3	duration	1080	0.191	10 / 0	0.000	0.179
BPIC15_3	handover	1080	0.616	0 / 0	–	0.565
BPIC15_4	duration	1080	0.623	0 / 0	–	0.715
BPIC15_4	handover	1080	0.618	0 / 0	–	0.784
BPIC15_5	duration	1080	0.328	0 / 0	–	0.507
BPIC17	duration	1080	0.353	25 / 0	0.000	0.203
BPIC19	duration	1620	0.360	38 / 24	0.632	0.563
Helpdesk	duration	1080	0.257	2 / 0	0.000	0.102
RoadFines	duration	1620	0.173	63 / 18	0.286	0.099
Sepsis	duration	1080	0.056	51 / 0	0.000	0.031
UCI498	duration	1080	0.493	17 / 17	1.000	0.623
UCI498	handover	1080	0.481	8 / 6	0.750	0.683

Table 6: The sign-disagreement rate of a one-number report, counted three ways, under the equal-level measure. The all-cells rate counts every admissible cell, most of whose signs the data do not determine; the resolved rate is restricted to the cells whose coverage-calibrated whole-surface band excludes zero, and the column before it gives the two counts it is computed from; the magnitude rate weights every cell by $|V|$ as well as by the measure. A pair with no resolved cell has no resolved rate and is printed with none rather than with zero.

with reading that as *what a one-number report exposes a reader to*, and both are corrected before the headline is stated.

The axes are not all the same kind of thing. The design space mixes choices an analyst makes with things that are not choices at all, and Table S13 partitions the decomposition accordingly: *analyst latitude* is the pipeline and the baseline rung; *resampling* is the split, five of whose six levels are expanding-origin folds on the same data; *counterfactual* is the register-quality condition, a different state of the world rather than a different way of analysing this one. On the primary scale, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, and resampling is the larger of the first two on 10 of the 19 pairs.

Whether the split moves the increment more than the analytic axes depends on which scale the partition is taken under, and we report both rather than the one that reads better. The primary scale decomposes *inside* each instrument and takes the median across the five, so the instrument is a stratum and its own contribution is in none of the three columns. Making it a sixth axis needs the headroom rescaling of Section 4.5, because raw AUC and raw average precision do not share units — and on that scale the ordering reverses: analyst latitude carries 19.2%, resampling 9.4% and the counterfactual 2.7%, with resampling the larger on 7 pairs. The instrument is an analyst's choice, so the second number is the one that answers the question as asked; the first is the one taken on the units the rest of Section 6 uses. Neither is a corpus fact independent of the scale, which is the paper's own thesis applied to the paper. Restricted to the 30 analyst-latitude cells a pair carries, the disagreement rate is 20.0% against the pooled 32.8%: the first is what a one-number report exposes a reader to, the second what a claim ranging also over degraded registers and resampled splits is exposed to.

The reference specification is itself a choice, and moving it moves the headline by less than the corrections above. The rate counts cells whose sign disagrees with the reference cell, so the reference supplies the sign and nothing else. Varying it one axis at a time over every level that axis carries — 26 variants in all, Table S14 — the corpus median runs 32.8% to 38.4% against the declared 32.8%: 35.3% if the reference pipeline becomes target-encoded boosting, and 32.8% to 38.2% across the five instruments. The headline is not an artefact of which cell we called conventional.

And most cells are not cells whose sign the data determine. The median pair resolves 8 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction, so a rate computed over unresolved cells is partly measuring that.

On the cells the corpus can resolve, the rate is 7.5%. Of the 896 cells whose coverage-calibrated whole-surface band excludes zero, 67 disagree in sign with the reference cell. Every cell it counts is one where the data do determine a sign, and that sign contradicts the number a conventional paper would print.

It is a pooled ratio, it depends on which band resolves the cells, and both facts belong beside it. The denominator is a sum over pairs whose resolved-cell counts run from zero to 332, not an average of per-pair rates, and it depends on which band decides what "resolved" means: the same quantity over the cells the *nominal* multiplier band resolves is 11.9%. The calibrated figure this paper quotes is the smaller of the two, so the choice is the conservative one — but it is a choice, and a reader is entitled to both. Under the empirical critical value of Section 4.3 the denominator is smaller again, 995 cells against 1,150. **And the disagreements are concentrated**:

of the 67 cells, 65 come from three logs, and only 4 of the 13 logs contribute any at all. Two of the three are also the pairs with the smallest bootstrap budget (Section 11). On 8 pairs no resolved cell contradicts the reference, on 5 some do, and on 6 nothing resolves.

Applying both corrections at once gives 12.9%, which is the number this section's own argument asks for. The two corrections above — restrict to the axes an analyst chooses, and restrict to the cells the band resolves — were made separately. Imposed together they leave 210 cells, of which 27 disagree. That is larger than the resolved-cell rate and smaller than the analyst-latitude rate, and it is the rate that answers "how often would a one-number report be wrong about a sign that the data determine, on a cell an analyst might actually have stood on".

Weighting by magnitude does not rescue the conventional report, and the rate depends on the measure. Weighting each cell by $|V|$ as well as by the measure gives a median 50.2%, *higher* than the unweighted rate: the cells that disagree are not the negligible ones. The median is 32.8% under the equal-level measure, 25.4% under the reference-weighted and 13.4% under the concentrated, falling as the measure concentrates on the reference specification, which is what it should do. A sign comparison is unit-free, which is why this one quantity may be pooled across instruments where an expected loss may not.

Table S11 attributes the disagreement to axes. Holding everything else fixed and changing one axis, the sign of the increment flips for a median 33.5% of same-everything-else pairs of cells under the split, 31.3% under the pipeline, 22.2% under the baseline rung, 20.4% under the register-quality condition and 19.7% under the metric. The two axes an evaluation paper is most likely to hold fixed without comment — which split, which pipeline — are the two that flip the sign most often.

6.6. The eight pairs where the construct is what this paper is about

Eight of the 19 pairs are IT service management, where the register is a maintained configuration or customer register and a data-quality mechanism has an operational referent. The other eleven include a loan amount treated as a high-cardinality categorical, a case-type reference nobody curates and an article of law, which Section 5.4 names as threats to construct validity. Table S10 reports every headline statistic on all nineteen pairs, on the eight, and on the eleven that remain, each with a cluster-bootstrap interval taking the *log* as the resampling unit — two targets on the same log share a cohort and are not two independent observations.

The higher-order share is 36.5% on the ITSM family against 56.0% on the whole corpus, the largest first-order index 35.5% against 28.7%, and the all-cells sign-disagreement rate 27.1% against 32.8%. One difference runs against the paper. **The headline is lower on the population this paper is about**: of the resolved cells, 3.5% disagree in sign on the ITSM family against 22.2% on the other eleven and 7.5% pooled. The corpus-wide figure is therefore driven by the pairs Section 5.4 names as carrying construct-validity threats, and a reader who takes this paper to be about maintained registers should read the ITSM column and should read it as the weaker case for the apparatus, not the stronger one.

6.7. Against specification-curve analysis as practised

The closest existing practice to what this paper proposes is specification-curve analysis [32]: enumerate the reasonable specifications, plot the curve, and test it against a sharp null by permutation using the median estimate and the share of the dominant sign. Table S15 runs that procedure unchanged on *every* admitted pair — 19 of them — over 100 permutations of a declared 24-cell sub-surface, beside the region label computed on the same cells, which makes "how often do the two verdicts part company" a measurement. A permutation replicate refits every cell of every pair, so the experiment carries two declared budgets rather than one: the sub-surface bounds the cells, and a cap of 20,000 cases per pair — a seeded sample, drawn once and restored to time order, so that the observed surface and its null are computed on the same cases — bounds the rows each fit sees. The cap binds on 7 of the 19 pairs and is recorded per pair in `results/s36_verdicts.csv`. The region labels this verdict is set beside are the ones of Section 6.4, computed on the full data.

They part company on 9 of the 19 pairs, which is close to half of them: the permutation test rejects the sharp null — the surface is decisively unlike one with no signal in it — and the same cells contain increments of *both* signs. A reader given only the permutation test on those pairs would conclude that the register matters and would have no way to learn that whether it helps or hurts depends on choices nobody told them about. On 3 of them the test does not reject and the region is unresolved: both instruments say the same thing and the cheaper one suffices. The rest are the mixed cases, where one instrument resolves and the other does not, and Table S15 prints all of them rather than the three that make the cleanest illustration.

And a specification curve is a statement about the cells it enumerates. On 1 of the 19 pairs every cell of the declared 24-cell sub-surface has one sign while the *full* surface for the same pair is sign-changing or conditionally harmful (Table 5). Nothing is wrong with either — they are claims about different sets — but it is a concrete demonstration that a specification curve without a declared denominator and a simultaneous band cannot distinguish "the sign holds" from "the sign holds on the cells I drew".

Neither procedure subsumes the other. The permutation test answers "is there anything here at all", costs one refit per permutation per cell, and needs no band; the region answers "may I quote a sign", costs a nested bootstrap, and is what a reader deciding whether to act on a number needs. This paper's contribution is the second, and the first should be reported beside it.

6.8. Decision rules, in sample and out

The four rules of Section 4.7 — *one-number*, *uniform*, *majority* and *weighted mean* — are compared under the loss defined there, inside one instrument at a time and in that instrument's own units. Table S20 and Supplement S10 carry it in full. The weighted mean is optimal in sample by construction, so what matters is out of sample: choosing on a random half of the admissible cells and paying on the other half, the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean, and is worse than the majority rule in 5 instruments of 5.

Three results run against the surface's advocate, and we report them rather than the comparison alone. The conservative rule is the worst of the four in 4 instruments, declining so often that the performance it forgoes exceeds the performance the one-number rule destroys; the median excess regret of the one-number rule is *zero* in every instrument, because on most pairs every rule picks the same action; and under rolling-origin folds the ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number. The case for reporting the surface rests on the decomposition and on the sign disagreement; this comparison is a check on those and not a result in its own right.

step	cohort	target	V at the knowledge rung	lo	hi	delta
the case study as published	reassignment	reassignment	-0.00322	–	–	–
the case study, tie-break declared	reassignment	reassignment	-0.00286	-0.00730	0.00203	0.00035
then the other tie-break rule	reassignment	reassignment	-0.00286	-0.00730	0.00203	0.00000
then the registered cohort	registered	reassignment	-0.00385	-0.00898	0.00069	-0.00098
then the registered target	registered	handover	0.00768	-0.00161	0.01217	0.01153

Table 7: The two published values of the register's increment at the later decision time, reconciled one factor at a time. Each row changes exactly one thing from the row above; 'delta' is what that change is worth. The first row is the number Section 7 reported and the last is the planned contrast PC3 of Table S8. Intervals are the pointwise basic construction at 400 draws; the first row has none because the ordering it was computed under is not reproducible by a rule, which is the subject of the second row.

7. Case study: a configuration management database

A configuration management database is the register an IT estate keeps of the items it contains, and it is bought partly on the argument that operational analytics need it. This section evaluates one against a named decision on a public log: will this case be reassigned to another group before it is resolved? The estate is a bank's and the log is BPI Challenge 2014 [11]; the register field is the affected configuration item, with 3,019 distinct values. It is reported at greater length than the other 19 pairs because it is the case that motivated the study and the one on which the decision time — when in the service-desk process the prediction is made — decides the answer.

7.1. One log, two published answers, and what separates them

This log appears twice in this paper. Section 6 applies the registered generic rules of Section 5, which admit 46,606 cases and define the target by the registered handover rule. This section uses the cohort and the reassignment target defined for the estate: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The first set is a strict subset of the second; the extra 1,151 cases are those the estate's own cutoff excludes.

The two produce different answers about the register at the later decision time, and the difference is large enough to change the sign. Against intake plus group plus the knowledge reference, the estate's cohort and reassignment target give -0.0029 and the registered cohort and handover target give $+0.00768$, on which the planned contrast PC3's one-sided p-value rejects at the Holm-adjusted level while its interval does not exclude zero (Section 4.4).

Table 7 takes the manuscript from the first to the second one factor at a time.

It is the target, not the cohort. Over the cohort $\times$ target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400; handover between activity groups, under the registered rule, has prevalence 0.927. They are not the same question and the register is not worth the same to them. Two further candidate explanations were checked and are not explanations: the fields reached by the two analyses agree on every row.

An axis nobody declares, and this paper was standing on it. Both analyses sort the log in time and split at seventy per cent of the row order, and 270 incidents share their timestamp with another. How those ties are broken decides which cases fall either side of the split, and the sort the case study originally used was not stable, so its tie order was neither declared nor reproducible: it printed -0.0032 where the two declared rules both give -0.0029. Over 25 random tie-breaks the increment at the knowledge rung ranges over 0.00088 AUC with a standard deviation of 0.00017, and it does not change sign (0 of 25 draws are positive), so no conclusion here turns on it — but it is a quarter of the point estimate's own magnitude, produced by a choice no reporting standard asks anyone to state. Every number in this section is computed under a declared tie-break, by incident identifier, on the estate's cohort and reassignment target, and every caption says so; the registered cohort's answers are in Section 6 with the other 19 pairs. The decision-time result below is therefore a statement about a routing decision on this estate, not about handover in general.

7.2. Three decision times

An incident in this estate does not begin as an incident. A caller reaches the service desk, an *interaction* is opened, the desk works it, and if it cannot be closed there an incident is created and routed. Three moments at which a prediction could be made are estimated, and what changes between them is not only which fields are admissible (Table 8).

τ_0, **first touch, every call.** The desk has just picked up the call and does not yet know whether it will escalate. The population is every interaction — 145,478 of them — the register is the interaction's own affected item, with 4,034 values, and the admissible baseline is the

decision time	baseline	n	prevalence	register levels	base auc	with auc	V	interval	resolved
T0 first touch, every call	intake	145478	0.146	4034	0.554	0.785	0.231	[+0.220, +0.262]	True
T1 first touch, escalating only	intake	44513	0.402	2445	0.552	0.754	0.201	[+0.193, +0.222]	True
T2 incident creation	intake	45455	0.400	2929	0.562	0.746	0.184	[+0.172, +0.204]	True
T2 incident creation	intake + group	45455	0.400	2929	0.644	0.748	0.103	[+0.095, +0.130]	True
T2 incident creation	intake + group + knowledge	45455	0.400	2929	0.805	0.802	-0.003	[-0.007, +0.002]	False
T2 incident creation, T1's population	intake	44513	0.402	2887	0.559	0.750	0.191	[+0.183, +0.213]	True

Table 8: The decision time implemented as an axis. Each moment carries its own population, its own register and its own admissible set; the PREDICTED TARGET is the same at all of them. τ_0 and τ_1 differ only in which cases are in scope. τ_1 and τ_2 differ in what is known AND in which cases are in scope, because 942 incidents have no interaction record; the fourth row is τ_2 restricted to τ_1's own population, and it is that row, not the third, that makes the τ_1-to-τ_2 step an information step alone. Every row is on the estate's own cohort and the reassignment target, under a declared tie-break, and every interval is the pointwise basic construction at 300 draws.

interaction's intake block. Against it the register is worth +0.2310 [+0.2204, +0.2618].

τ_1**, first touch, the calls that escalate.** The same moment and the same fields, restricted to the 44,513 calls that did become incidents. The register is worth +0.2013 [+0.1930, +0.2224].

τ_2**, incident creation.** The desk has worked the interaction and escalated it. The population is the 45,455 incidents; the group that logged the incident is known, and so — probably — is the knowledge-article reference the desk consulted. Against the intake block the register is worth +0.1838; against intake plus group, +0.1034 [+0.0763, +0.1115]; against intake plus group plus the knowledge reference, −0.0029 [−0.0078, +0.0015] — the one rung of the five that does not resolve.

The pairwise differences separate the mechanisms that a baseline ladder alone conflates, and separating them properly requires one more row than the three moments give. τ_1 and τ_2 are *not* the same cases: 942 incidents have no interaction record, so τ_2 carries 45,455 cases against τ_1's 44,513. Table 8 therefore also reports τ_2 restricted to τ_1's own population — the same 44,513 cases at the same prevalence 0.402 — which is the row that makes the information step an information step.

Restricting the population from every call to the calls that escalate, holding the information fixed, moves the increment by −0.0297 AUC. Moving from first touch to incident creation *on the matched cases*, changing what is known and nothing else, moves it a further −0.0099. The remaining −0.0076 is what the 942 incidents with no interaction record are worth, and it is a population difference rather than an information one: the unadjusted τ_1-to-τ_2 step, -0.0175, is the sum of the two and attributing all of it to information would overstate the information channel by three quarters. Admitting the group and then the knowledge reference takes the increment the rest of the way to zero. A decision time is therefore an argument of the estimand in its own right: it fixes the population, the feature snapshot and the admissible set, and only the third of those is a baseline.

Neither τ_0 nor τ_2 is “the” answer: which one a reader wants depends on where in their own process they intend to predict. A desk screening at τ_0 looks at 145,478 calls a year at a prevalence of 0.146; a desk screening at τ_2 looks at 45,455 incidents at 0.400, and Section 8.4 prices the difference.

7.3. The leakage tipping point

The knowledge reference is load-bearing: it is the field that takes the register's increment to indistinguishable from zero, so the evidence that it was available at τ_2 is stated exactly, and it is weaker than proof. The interaction detail file establishes that the field is written by the service-desk process and not by the incident process — 94,250 interactions in it never became incidents, and 100% of them carry a knowledge reference — but not *when*, for a given incident, the value was written. Table S23 lists each piece of evidence beside what it does and does not establish. Availability at τ_2 is probable and unproven, and the disposition of an unprovable admissibility assumption is a sensitivity analysis rather than a headline.

Because availability is probable rather than proven, the increment is reported in Figure 4 as a function of λ, the share of knowledge references assumed to have been written after the incident existed and therefore inadmissible. Masking a random λ share of the field and re-estimating traces a curve from the permissive end ($\lambda = 0$, the field fully admissible) to the conservative end ($\lambda = 1$, equivalent to τ_2 without the field at all). The masking is stochastic, so each point is a mean over 5 seeds with the spread across them printed beside it, and the curve is a sensitivity display rather than an estimate the paper quotes a number from.

At $\lambda = 0$ the register is worth -0.0029 [-0.0078, $+0.0015$], which is not resolvably different from zero, and it reaches half of its knowledge-free value at $\lambda = 0.64$: a reader who believes that more than 64% of knowledge references are written *after* the incident exists should read this estate's register as worth at least half of $+0.1034$ AUC, and one who believes almost none are should

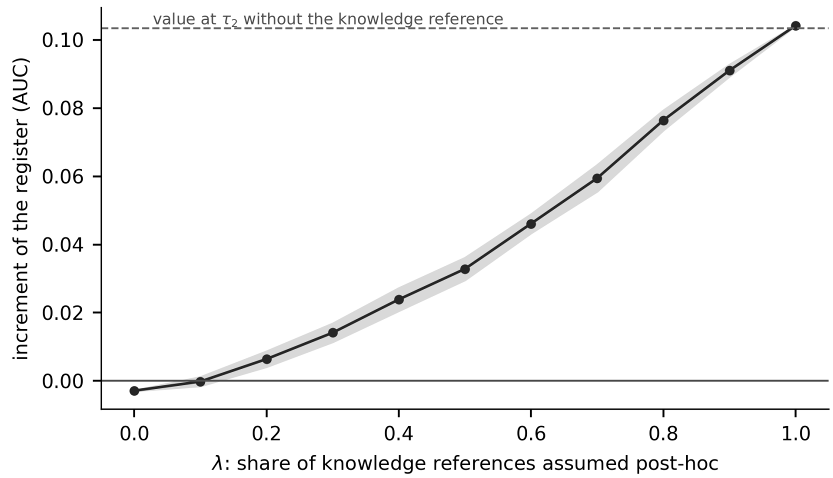

Figure 4: The register's increment at τ_2 on the estate's cohort and reassignment target, as a function of λ, the share of knowledge-article references assumed to have been written after the incident existed. A reader who has a belief about λ can read their own answer off the curve.

read it as worth almost nothing. We have no way to measure λ and do not claim one; the curve makes the assumption explicit and continuous rather than binary and made on the reader's behalf.

7.4. Register quality

Masking a complete field is a sensitivity experiment, not a model of a real register, so Table S24 reports the six mechanisms of Section 3 plus a discovery lag, and sweeps the accuracy and reconciliation mechanisms rather than reporting one level of each. Everything here is on the estate's cohort and reassignment target; Table S25 carries the severity curves, over 11 severities and 20 seeds per stochastic mechanism, and Table S27 the same mechanisms on the registered cohort. Four things follow, and two of them are nulls we did not expect.

Coverage is the expensive failure, and losing the long tail is not the worst way to lose it. At half population the register is worth +0.0819 if its long tail is absent and +0.0708 if its core is, against +0.1034 complete. Losing the core costs more than losing the tail, which is the opposite of the usual intuition that the rare identities carry the information: on this estate the frequent items are the ones the model has enough cases to learn from.

Loss at random is worse than either (+0.0676), because it fragments every identity rather than removing some cleanly.

Accuracy costs about what coverage costs. Corrupting three tenths of the identities — the register is complete, and wrong — leaves it worth +0.0689, a loss of −0.0346, about what losing a third of its population costs. A maturity assessment that counts populated rows and does not sample them for correctness is measuring one of the two and reporting it as the whole.

Reconciliation failure is nearly free. Splitting *every* identity into two records moves the increment by −0.0019: with thousands of cases per identity, halving the evidence behind each still leaves enough.

Staleness is exactly zero, and the reason is the encoder. Removing identities the register had not seen by the split point moves the increment by +0.0000. That is not a bug in the mechanism: a one-hot encoder fitted on the training half already ignores levels it has never seen, so a test row carrying an undiscovered identity contributes an all-zero encoding either way. Under an encoder that keeps missingness as an explicit indicator the effect is non-zero; Supplement S11 measures that on the registered cohort. A *discovery lag* — identities first used in the last half of the training window absent everywhere, leaving 94.0% of rows populated — moves it by +0.0039. Whether a data-quality defect is visible is therefore an interaction between the quality axis and the encoding axis, and an analyst who varies one without the other will conclude that the defect does not matter.

And which rows go missing matters as much as how many. The three *dependent* mechanisms of Section 3 carry severity integrals +0.0768, +0.0486 and +0.0559, against +0.0692 for the independent long-tail mechanism, which the dependent ones bracket rather than track: the identity of the missing rows changes what the loss costs by up to half again, so a quality axis built only from independent coin flips is measuring one corner of the question. The summary is an integral against a uniform measure on severity, and coarsening the sweep from 11 severities to five moves it by at most 0.0114 AUC. Every mechanism is verified by execution to be a function of the training half alone — 16 checks run and 16 pass.

7.5. What this says to a configuration management programme

The register in this log is layered, and the layers are the ones a common service data model uses: an item belongs to a class, a class to a subtype, and a subtype rolls up to a service component. Supplement S4 estimates the increment at each layer against the intake block, on the registered targets.

The coarse layers are *not* free — on the handover target the item's type alone is worth +0.134 and its subtype +0.163, against +0.257 for the item itself, and on duration +0.045 against +0.129 — so knowing that a case concerns *a laptop* is worth between a third and two thirds of knowing which one. What the item adds *on top of* the subtype is the quantity a populated hierarchy does not buy, and it is +0.109 on handover and +0.087 on duration. A programme that populates the class hierarchy and stops has bought a real part of the value and left the larger marginal part on the table; it has not bought nothing. These figures are on the registered targets rather than this section's reassignment target, and Supplement S4 says so.

The three dependent quality mechanisms map onto how a configuration management database actually decays, which is why they are in the design space rather than independent coin flips. *Discovery lag* is the interval between a service going live and the discovery tooling finding its items, which is why a new service's cases carry an empty affected item for weeks. *Unreconciled duplicates* arise when two discovery sources write the same physical item under two identifiers. *Propensity-dependent missingness* arises when the affected item is populated by the desk rather than by discovery: the unusual case, which is the one worth predicting, is the one the desk is least likely to classify correctly under time pressure. The measured costs above say which of the three is worth spending on: on this estate, coverage and accuracy at the item level, and not reconciliation.

Two disclaimers. This is predictive performance on one estate for one decision, not procurement advice, and a register that pays nothing here may be bought for change control, audit or impact analysis, none of which this measures; and the ranking of the mechanisms is a property of this cohort's cardinality and volume.

7.6. The absorption, and why the ratio is secondary

The quantity a report of this kind usually leads with is the *relative* reduction R: the share of the register's apparent value that a single free field absorbs. The absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ is reported first, with its own interval, and R second with a Fieller set and the set's kind. On this estate at the reference cell D is +0.0804 [+0.0626, +0.0914] and R is 0.437 with a bounded Fieller set; 0 of the 30 reductions this ladder produces have a Fieller set that is not a bounded interval, and for those a percentile interval is not a summary of anything. Across the corpus the figure is 5,282 of 10,584, and the full ladder is Table S26.

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit, which is: the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. The operating point is a cost ratio, not a knob

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\mathrm{NB}(\theta) \;=\; \frac{\mathrm{TP}(\theta)}{n} \;-\; \frac{\mathrm{FP}(\theta)}{n} \cdot \frac{\theta}{1-\theta},$$

so $(1-\theta)/\theta$ is the number of false nominations the desk is willing to accept for one true one. Choosing θ is choosing an operational cost ratio, and it belongs to the analyst's declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$, or 9.0 false nominations accepted per true one, where a review is cheap and the desk nominates liberally; 0.30, or 2.3, where a review and a miss cost about the same; and 0.60, or 0.7, where review capacity is scarce. A reader whose desk sits at a ratio not in that list reads their own point off Figure 5: the operating point is the reader's to choose, so the analyst reports the curve rather than a point on it.

8.2. Calibration comes first, or the threshold means nothing

The identity above reads θ as a probability threshold, and everything that follows from it requires the score to be a probability. A threshold of 0.30 applied to a score whose calibration slope is -0.28 is not a statement about a cost ratio of 2.3 false nominations per true one; it is a statement about a number that happens to lie at 0.30 on an arbitrary scale. Every model contributing to a decision-curve conclusion in this paper is therefore refitted under a **training-only** calibration: isotonic and Platt scaling, each cross-validated *inside* the training half, so no test outcome touches the calibrator, and the calibration is rebuilt inside every resampling iteration. Table S22 gives the diagnostics for the raw and the calibrated scores on every pair.

The correction bites. Across the corpus the median calibration slope is 0.63 on raw scores and 0.95 after isotonic calibration; the median expected calibration error falls from 0.105 to 0.089; and the number of arms whose slope lies outside $[0.5, 2]$ falls from 92 to 63. It also changes conclusions: the

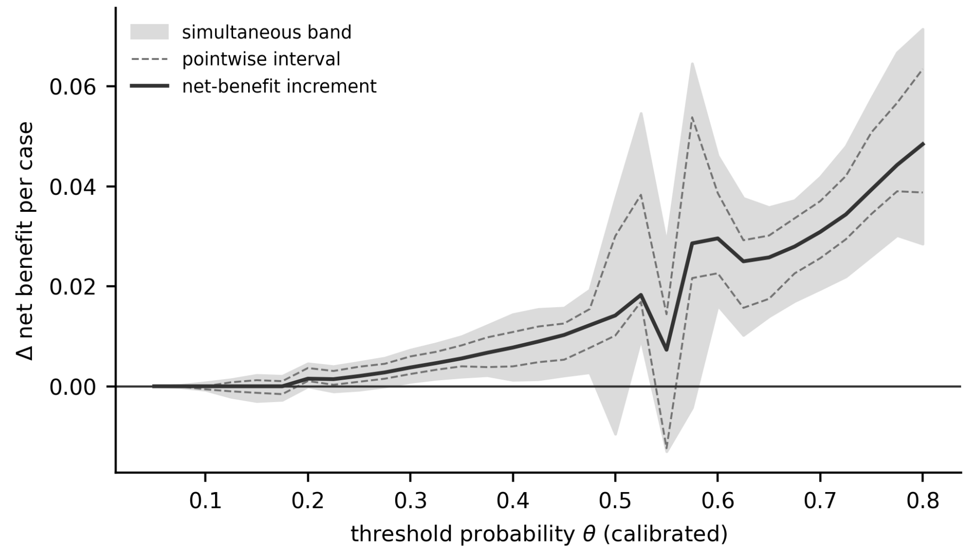

Figure 5: The register's net-benefit increment across the operating range at the reference cell, with the pointwise interval and the simultaneous max-t band. The simultaneous band is what a claim about a *range* of thresholds has to be read against. The curve dips near $\theta = 0.55$, where the simultaneous band's lower edge reaches -0.0130: 0 of the 31 operating points resolve as harmful pointwise and 0 under the simultaneous band, so the dip a reader can see is not a harm the evidence resolves.

sign of the net-benefit increment differs between the raw and the calibrated curve at a mean 27.4% of thresholds.

What may and may not be read as a decision curve. Curves computed on calibrated probabilities carry the probability-threshold interpretation and are what Section 8.3 and the case study read. Curves on raw scores are reported as *score-threshold sensitivity analyses* and are labelled so in every table; they say how a conclusion moves as a selection rule tightens, and they do not say what cost ratio that rule encodes.

8.3. Simultaneous bands change what can be said

Figure 5 draws the register's net-benefit increment across the grid at the reference cell on *calibrated* probabilities, with the pointwise interval and the simultaneous max-t band. The band's family is declared separately from the scalar family and spans every admissible net-benefit cell of the pair — 248 members, critical value 3.57 — and the difference from a pointwise reading is not cosmetic. Its median width is 0.0136 against the pointwise interval's 0.0071, and across the 31 operating points of this estate's decision curve the pointwise interval declares the increment resolvably positive at 24 of them while the simultaneous band declares it at 18. Both are built from the same draws, so the only thing that differs is the family: those points are the price of a statement about a *range* of thresholds rather than about each threshold separately, and a paper that reads a decision curve across its range and prices it pointwise has claimed the first while paying for the second.

The curve dips near $\theta = 0.55$, where the simultaneous band's lower edge reaches -0.0130, so the count of points that resolve as *harmful* is reported rather than left to the eye: 0 of the 31 points pointwise and 0 under the simultaneous band. The second can only be smaller than or equal to the first — the wider family gives the wider interval at every point, which `verify_numbers` checks — so a dip a reader can see is not a harm the evidence resolves. Over the 10,974 decision-curve cells the corpus carries, the same counts are 3 and 1; Table S18 prints this pair's own curve, operating point by operating point, and its caption names the cell it is computed at.

And this family is the one most exposed to how the critical value was estimated. The decision-curve bands are *not* coverage-calibrated: the (n, K) calibration of Section 6.4 is estimated for the scalar instruments and is not applied here. On every one of the 19 decision-curve families the multiplier critical value lies below the whole order-statistic interval of the empirical quantile (Section 4.3), by a median factor of 2.09, and across the corpus's decision-curve cells the multiplier band resolves 3,066 against the empirical band's 613. A reader who takes the empirical value should read this section's counts as upper bounds. Section 11 states why we do not choose between the two.

8.4. The decision, in units a service desk uses

Because net benefit is a single cardinal utility, the decision-rule comparison of Section 6.8 can be run on it without the unit problem that forces that section to work instrument by instrument. Three things have to be settled first, and each is settled by a declared rule rather than a default.

Which models may enter. A threshold is a cost ratio only if the score is a probability, so a model whose recalibrated slope lies outside $[0.5, 2.0]$ is excluded from the decision-analytic reading by a registered rule applied at the level of the (pair, pipeline, baseline) triple and requiring both arms to pass. A *model* here is one such triple, not one fitted classifier. The rule removes 40 of 118 of them and leaves 3 of 19 pairs with none. **That count is itself the strongest operational finding in this section**: on a third of the models a published evaluation would fit here, no threshold can be read as a cost ratio at all.

Which thresholds. A uniform measure over an operating range from 0.05 to 0.80 is not a desk's belief. For a decision about pre-empting a reassignment the false positive costs a specialist's few minutes and the false negative costs a re-triage, a second queue wait and the service-level exposure that goes with it; ratios between three and ten to one put the indifference threshold between 0.08 and 0.36, with a median of 0.19. A Beta distribution with that support is declared as the *desk distribution*, and single exchange rates are reported beside it.

Which unit. Net benefit per case is a rate near zero, and three decimal places of it look like a tie, so Table S16 reports it per thousand cases.

And the rules still mostly tie. Under the desk distribution the one-number rule, the majority rule and the weighted mean all carry a median excess regret of 0.000 per thousand cases; only the conservative rule loses, at 0.048. The four rules do not all choose the same action on 11 pairs, and where they differ the one-number rule is the worst on 10 of them, at a maximum of 0.412 per thousand — a real ordering and a small quantity. **The case against a one-number report does not rest on this design.** It rests on the decomposition of Section 4.5 and on the sign disagreement of Section 6.5, which is unit-free.

What does separate, in this unit, is the decision time. The register is worth 1.71 true positives per thousand cases against the intake block. Against intake plus group plus the knowledge reference — the same estate, the same register, a later moment in the same process — it is worth 0.003. A desk that read the first number and deployed where the second applies is short 1.70 per thousand cases. That is the operational content of Section 7.2, and it is larger than every difference between the four collapse rules put together. It is also conditional on the knowledge reference being admissible at τ_2, which Section 7 shows is probable and unproven: the second figure is the $\lambda = 0$ end of Figure 4, and a reader who believes a share λ of those references are written after the incident exists should read their own figure off that curve.

What the operational statement does not say. Net benefit converts a discrimination increment into units the desk recognises. It does not convert it into money, and three gaps remain, each needing data this study does not have: nominating a case for review is not the same as preventing a reassignment, and whether acting on a nomination changes the outcome is an interventional question on observational evidence; the exchange rate is declared rather than measured, which is why the whole curve is reported; and the cost of the register itself — discovery tooling, reconciliation effort, the staff who keep it current — is nowhere in this evidence.

9. A reporting standard, and software

9.1. The minimum reportable form

The apparatus of Sections 3 and 4 implies a reporting standard short enough to state in full. A feature-value claim is reportable when it carries: **the design space**, as an enumerable declaration of which learners, encodings, targets, splits, decision times, quality mechanisms, baselines, metrics and operating points are admissible, why anything defensible was left out, and *the measure placed on those levels*; **the denominator**, as a count a reader can reproduce from the axis levels — declared cells, computed cells, and the cells any inference actually rests on, which need not be the same set; **the absolute increments** at every rung of the baseline ladder, in the metric's own units, with intervals from a resampling scheme that refits the model; **the sensitivity decomposition**, with the total higher-order share reported as $1 - \sum_i S_i$ and not as any per-axis difference; **the resolution region and** ρ, under a band whose family is the set the label ranges over; and **the ratio** R **last if at all**, with a Fieller set and the set's kind, never where the denominator is not resolvably positive. Nothing in that list is expensive once the surface exists, and the surface is a factorial over declarations the analyst has already made implicitly.

9.2. `fieldvalue`

The standard ships as `fieldvalue`, which computes the surface from a data frame, a feature name and a dictionary of baselines. Its notable property is a refusal: `Surface` raises rather than returning a scalar, `Surface.report()` emits the form above, and `Surface.at(...)` returns one cell only if the caller names every axis. Version 0.2.0 adds the four objects of Sections 4.5–4.7 as functions of a surface — `decompose`, `regions`, `robustness`, `regret` — each taking a tidy frame rather than this package's own object, so they apply to a surface computed by some other pipeline. It carries 108 tests and is an *independent implementation*: the analysis scripts and the package compute the same four objects from the same definitions in different code, neither written against the other, and on this corpus they agree on 266 of 266 quantities to 1.1×10^{-16}.

9.3. Beyond process logs, and beyond this paper's claims

The standard is not about event logs and not about configuration databases. Supplement S7 runs it on UCI Adult, where the expensive field is `occupation` and the cheap ones come with the sampling frame: over 480 cells the reduction runs 0.200 to 0.947; the baseline and the metric carry the same share of the variance to three figures, 25.9% and 25.9%, against 7.2% for the register population; and a one-number report misstates the sign for a mean 24.9% of the other cells. `examples/worked_example.py` produces all of it in 121 statements against the package's public interface.

How the surface is reported today. Whether the axes this paper varies are varied in published work deserves a systematic review, and we did not conduct one. A machine-assisted prevalence pilot was run on a convenience frame and is *not* reported here: with one adjudicator, no independent human reliability study and a mechanical screen whose measured sensitivity is 0.474, it can generate a hypothesis about practice and cannot test one. Its frame, sample, codes, adjudications and evidence dossiers are in the archive. The paper's argument does not need it: the case against one-number reporting is made by the decomposition and the sign disagreement, on data, and not by a count of how often other authors report one number.

10. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the estimator where it is. The full tables are in Supplement S11 and the diagnosis of the one world that failed is in Supplement S6.

10.1. Six worlds, two estimands

A simulation that fits a logistic data-generating process with logistic regression tests an unusually favourable arrangement. This one uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each. The covariate space is finite, so the joint distribution of cell and outcome is enumerated and the population AUC of any scoring rule is computed in closed form rather than simulated. The worlds are a logistic surface, a nonlinear one with an interaction and a threshold the logistic model cannot see, a drifting one, a sparse register with 200 levels, an imbalanced outcome at prevalence 0.02, and a noisy register in which a share of values are wrong.

Two estimands are separated because under misspecification they differ. V_{oracle} is the increment between the two Bayes-optimal scoring rules — what the register is worth in the world. V_{limit} is the increment between the two fitted pipelines' limits — what the estimator estimates. A resampling interval can only cover the second, and reporting its coverage of the first as though it were the estimator's coverage is a category error. Both are reported in Table S30. Where a register value is corrupted the cells the generator writes down are not the cells the estimator observes, so the truth has to be marginalised over the corruption; the two experiments that found that and the check that guards it are in Supplement S6.

10.2. Coverage, bias and width

Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points at 1,000 replicates, which is the resolution every comparison here should be read at; Figure S2 draws them by construction and world.

Where the refit helps. The nested basic interval covers at least as well as the fixed-model percentile interval on 6 of the 6 worlds, and its median across them is 93.0% against the fixed-model interval's 92.1%. Both fall a little short of nominal on most worlds, by more than the Monte Carlo resolution.

Where the percentile interval fails, and why. On the sparse world it collapses to 37.2%, and the *nested* percentile interval is worse there than the fixed-model one. The cause is Section 4.1: a bootstrap training resample holds about 63% of the distinct register levels, so every refit sees a sparser register and the bootstrap distribution shifts below the estimate. The basic interval pivots that shift out and restores coverage — 89.9% — and costs nothing where there was nothing to repair, at a median 93.0% and a minimum 89.9% across all six worlds. The bias-corrected percentile does *not* repair the sparse world (63.1%), which is why every interval and band in this paper uses the basic construction. The m-out-of-n subsampling interval, the standard alternative in exactly this regime, does not win either: 72.8% on the sparse world and a median 75.8% across the 3 worlds it was priced on.

What no interval repairs. Against V_{oracle} rather than V_{limit}, every construction undercovers on the drifting world — 65.8% at best — and it should: when the coefficients move between eras the estimator is not consistent for the oracle quantity, and the largest gap is 0.0161 in AUC units. Under misspecification an interval around the estimator's own limit is not an interval around the quantity a reader cares about, and no resampling scheme repairs that.

10.3. Sample size, register cardinality and block length

A construction that works at one sample size, one register cardinality and one block length has been validated at a point, not in a regime. Sample size and register cardinality are therefore varied *jointly*, up to $K = 3{,}019$ levels — the cardinality of the case study's own register — and the block length is varied around its rule of thumb.

The ratio K/n captures most of the dependence, and not all of it. Across the plane the basic interval covers a median 83.3% and a minimum 21.3%, the last at a ratio of 0.377. Below $K/n = 0.050$ its coverage lies between 88.7% and 90.7% — short of nominal everywhere, by about one Monte Carlo standard error at 150 replicates — and it has fallen below three quarters by 0.125. The ratio is the right first-order summary — the mechanism of Section 4.1 is why, and varying n alone would not have exposed the regime the corpus occupies — but it is not sufficient, and the plane says so in two places. The two cells closest to each other in ratio — a register of 200 levels at 2,000 training rows, and one of 1,000 at 8,000 — cover 0.780 and 0.613, a gap of 3.2 combined Monte Carlo standard errors; and holding the register at 1,000 levels while moving the training half from 2,000 rows to 8,000 moves coverage from 0.453 to 0.613, so n carries signal of its own. The calibration of Section 6.4 is a function of the ratio alone and therefore inherits this residual, which Section 11 states.

Where this corpus sits, and what is done about it. The median log–target pair here has $K/n = 0.106$, the range runs 0.001 to 0.324, and 10 of 19 pairs are above a tenth. Most of this corpus therefore sits at or beyond the ratio at which coverage starts to fall, and a nominal band on those pairs is *narrower* than a 95% interval should be — the anti-conservative direction, which inflates the count of resolved cells. The plane is dense enough to calibrate against, and Section 6.4 does.

Block length. 3 lengths are examined — half the rule $\lceil n^{1/3} \rceil$, the rule, and twice it. Coverage moves with the length, spanning 3.6% within a world, of the order of two standard errors; but the comparison the paper rests on is unchanged at every length, the basic interval beating the percentile interval on the sparse register by at least 46.4% at all 3.

10.4. The band's family-wise coverage

Everything above validates a *pointwise* interval. Two of this paper's four reporting objects are functions of a *simultaneous* band, and until this section nothing measured what that band covers.

The design, and why its answer is an upper bound. The truth is zero at every cell of a synthetic family. One draw from the sampling distribution gives the point estimate, and as many further draws *from that same distribution* as the corpus family it is matched to carries give the bootstrap — so the bootstrap is not an approximation of the sampling distribution but is it. Every way a resampling scheme can be wrong — dependence it fails to reproduce, a refit that comes apart, a block length misjudged — is switched off, and what is left is the object under test: a standard error and a critical value both estimated from the same handful of draws. A coverage measured here is therefore the most the real procedure can attain, and a shortfall here is one that no improvement to the resampling repairs. The families are matched to this corpus in size, draw count, cross-cell correlation, effective rank and per-cell excess kurtosis; the matching is fitted by grid search against the corpus's own draw files and reported beside its target rather than asserted. Coverage is over 2,000 replicates in each of 18 cells, so no coverage here is determined worse than 1.1%.

Three regimes, because "the draws are heavy-tailed" turned out to name two different things. Under *Gaussian* draws the multiplier's family-wise coverage is a median 87.3% and a minimum 74.8%. Under draws carrying this corpus's tails, whose median cell has an excess kurtosis of 0.32 on the surface families and whose ninetieth percentile is 3.6, the median is 84.6% and the minimum 76.1%. Under the third regime, which adds the degenerate cells described below, they are 84.4% and 39.2%.

What is wrong with the decision-curve families is not a tail. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic rather than by estimation, and 841 cells of the corpus's curve families are exactly zero in at least nine draws of ten — up to 17.2% of a single family, against 0 such cells anywhere in the surface families. The admission rule keeps any cell whose spread exceeds a floor in absolute units, so such a cell survives and standardising it by that spread sends its studentised value to the largest value a deviate standardised over B points can take. They are what makes q_{emp} large on those families: excluding them moves the median ratio q_{emp}/q from 2.09 to 1.80. It does *not* dissolve the disagreement — q still lies below the empirical interval on 19 of the 19 curve families and 15 of the 19 surface ones once they are gone — and Table S33 reports the regime with and without them because Section 4.3's count of what q_{emp} resolves is, on those families, partly a count of arithmetic.

How far short, in the units the calibration works in. A critical value is widened multiplicatively, so the useful summary of the shortfall is the factor that would have closed it. On the surface families that factor is 1.05 to 1.24; on the decision-curve families, 1.90 to 2.05. The coverage calibration of Section 6.4 applies 1.09 to 1.88, is fitted to restore *pointwise* coverage rather than this one, and is applied to the surface families only — so it is of the right order where it acts and absent where the shortfall is largest. Section 11 says what follows.

The draw count is the binding constraint. At the modal surface family the multiplier band covers 82.0% at 33 draws and 96.6% at 1,000, and the per-cell interval underneath it moves 90.3% to 94.9% over the same range: part of what the band loses it loses before any critical value is chosen, and Section 10.2 validated that construction at a refit count this corpus does not reach on most pairs. No estimator recovers the difference. The best of the five, the empirical quantile at the upper end of its own order-statistic interval, reaches 92.2% at 1.52 times the width and still falls to 86.5% at the smallest family; a Rademacher wild multiplier, which preserves the observed draws' magnitudes instead of replacing them by Gaussians and is the natural repair for a tail problem, is worse than the Gaussian multiplier at 80.7%.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a plane of 10 (n, K) cells and 3 block lengths, and the band on the 18 matched families of Section 10.4 — on nothing else. The stationarity and mixing the moving-block bootstrap needs are assumed and not tested; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about.

11. Limitations

This is predictive performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a prediction changes an outcome, or what a data programme is worth. The quantity is an increment in predictive performance, read at a declared exchange rate in Section 8. A reader who converts Table 3 into a procurement decision is doing something this evidence does not support.

The design space is declared, not complete, and one omission is structural. $\mathcal{S}$ has nine axes here. It could have more — hyperparameter tuning protocol, missing-data treatment for the baseline fields, the horizon of the target, the label-noise model — and the cohort rule is varied only on the case study's log, so the honest form of the claim is "the increment moves this much over *this* space". The structural omission is that **prefix length, prefix bucketing and sequence encoding are not axes here and cannot be**, because this study predicts at one prediction point from creation-time attributes. Those are the axes predictive process monitoring varies most, so the corpus result measures sensitivity in a simpler adjacent setting on the same data rather than in that literature's own benchmark, and adding a prefix axis on at least one log is the clearest next study this paper implies.

The axes are not all the same kind of thing. Three of the nine are choices an analyst makes, one is a resampling scheme and one is a counterfactual state of the register, and a variance decomposition that pools them answers a question nobody asked. Section 6.5 partitions them and reports the sign-disagreement rate on the analyst-latitude sub-space (20.0%) beside the pooled one (32.8%). The region labels and ρ are *not* partitioned, and are therefore conditional on the quality axis being averaged over: a surface is asked to be resolvably beneficial in cells where three quarters of the register has been deleted, which is part of why no surface here is uniformly beneficial.

The inference surface is smaller than the declared surface, and the grid is not uniform across the corpus. Every point estimate comes from the full declared space; every *band* comes from the inference surface, a median 12.5% of each pair's computational surface. It is a subset of the declared surface cell for cell and the share of positive cells differs between them by a median 0.067, but a study with compute to spare should equate the two. The same budget shows in two further places. The draw counts run 40 to 150 as declared and 33 to 150 effectively, because a replicate that leaves any cell of a family without outcome variation is dropped from that family; the two families that lose the most sit on logs that also supply much of the sign disagreement of Section 6.5, so the headline's largest contributors are also its noisiest. And the pipeline axis carries four levels on the case study's log and two or three elsewhere, by a rule declared on row count before the runs, so a sensitivity index computed on one pair is not strictly comparable with one computed on another.

Every weighted quantity depends on a measure that is a modelling choice. A decomposition, a robustness index and a regret are all taken with respect to a distribution over specifications, and there is no neutral one. Three product measures and an envelope over a fourth are reported (Table S19); across the three the total higher-order share moves by 24.3%. **One qualitative conclusion does not survive all four**: under a measure concentrated on the reference specification, a single axis carries more variance than everything higher-order combined on 12 of 19 pairs, so "the interactions dominate" is a claim about a measure and not only about a surface. Two structural invariances, to duplicating a level and to refining a grid, are enforced and tested, so no number here can be moved by re-describing the same design space. Separately, the decomposition inherits the metric's scale (Proposition 3), which is why the within-instrument decomposition is primary and the cross-instrument one is a sensitivity analysis.

One estate for the case study, and a heterogeneous corpus. The configuration management database analysis is one bank's estate, on one public log, and the availability question at its centre is settled by probability rather than by a timestamped field-value history; the strongest version of this study would have an organisational partner supplying timestamped field histories, a known prediction time, historical register population and accuracy, and operational review costs. The 19 pairs are also not replications of one another: across the 13 logs the two registered targets label the same case identically on a median 51% of cases. Every corpus-level statistic is therefore about *sensitivity*, and every one is reported on the ITSM family as well as on the whole corpus (Table S10).

The quality mechanisms are synthetic. Population loss, identity error, reconciliation failure and staleness are constructed from the training half, not observed. Three of them are *dependent* — on time, on the item's recorded type, and on a training-estimated propensity that correlates missingness with the outcome — which is closer to how a register degrades than an independent coin flip, and every one is verified by execution to read no test outcome. But a constructed mechanism is not an observed one, and validating them against a genuinely incomplete register is the obvious next study.

The coverage calibration is no longer an extrapolation in n, and what replaced that problem is worse. The plane that identifies the inflation factor is estimated at training sizes 500 to 180,000 against a corpus running 735 to 176,213, so every one of the 19 pairs now lies inside the range the factor is identified on. Two things took the place of the extrapolation.

The applied curve is misspecified, measurably. $\log n$ enters the fit at $t = 8.3$ (Section 6.4): at a fixed ratio the widening a pair needs grows with its sample size, which Section 10.3 could assert before and can now measure. The curve the corpus is calibrated with is a function of the ratio alone and therefore carries that dependence as a residual — it sits below the measured factor's own lower confidence limit on 2 of the plane's cells, by at most 0.18, both of them at large n. Putting $\log n$ into a parametric curve does not repair it and makes the object worse on the criterion the monotone curve was chosen under (Supplement S9). Every calibrated region label and every ρ on a large pair is therefore corrected by slightly less than the plane says it needs.

On 4 of the plane's 33 cells the factor does not exist. The inflation factor widens an interval about its point estimate, and past a crossing in n the basic interval no longer contains that estimate (Section 4.1), so there is nothing to widen about. All 4 are at 50,000 training rows or more. **This is not confined to the simulation**: on the corpus itself 74 of the 3,900 whole-surface cells have a band that excludes their own point estimate, on 3 pairs and reaching 21.1% of the cells on the worst. For those cells the calibrated band is a widening of an interval the calibration's own estimator could not have been fitted on, and we do not know what it covers.

The interval construction was chosen on evidence from one sample size, below the size at which it misbehaves. Section 10.2 compares seven constructions and selects the basic one, and that comparison runs at 4,000 training rows and at no other size. Section 4.1 reports that the same construction produces a band excluding its own point estimate once the bootstrap displacement exceeds the band's half-width, which the plane places above 8,000 rows at a ratio of 0.125 — so the evidence that chose the construction was gathered below the regime in which the construction develops the defect. Whether the percentile, the bias-corrected or the m-out-of-n interval behaves better up there is not known: none of them was priced above 4,000 rows either. Re-running that comparison across the plane's range in n is the first thing we would do, and it is cheap relative to the surface.

The band's critical value is estimated twice and the two estimates disagree. This is the sharpest limitation in the paper and it is not one the calibration of Section 6.4 addresses. The multiplier bootstrap's value lies below the entire order-statistic interval of the empirical quantile of the same observed maxima on 15 of 19 surface families and on every one of the 19 decision-curve families, in the anti-conservative direction, because the draws are heavier-tailed than the Gaussian process the multiplier simulates. Section 4.3 reports what each estimator resolves — 1,150 surface cells against 995, and 3,066 decision-curve cells against 613 — and a reader who prefers the empirical value should read the second of each pair.

The band does not attain its nominal level, and we now know by how much. Section 10.4 measures family-wise coverage against a known answer over families matched to this corpus, and the multiplier band covers a median 84.6% and a minimum 76.1% against a nominal 95% — 87.3% and 74.8% even under Gaussian draws, which is the case the construction is derived for. The bootstrap in that experiment is *ideal*, so those are upper bounds on what the procedure attains here.

What that leaves the two families is not the same, and the difference is the calibration. Section 10.4 also reports the multiplicative widening that *would* have given nominal coverage on each matched cell, which is directly comparable with the inflation factor of Section 6.4 because both widen the same critical value. On the surface families that widening is 1.05 to 1.24; the calibration already applies 1.09 to 1.88, a median 1.29, so on those families it is of the right order and may already absorb the shortfall — it was fitted to restore *pointwise* coverage and not this, so that it does is fortunate and not by design, and we claim no more than the order. On the decision-curve families the widening needed is 1.90 to 2.05 — about a doubling — **and the calibration does not touch them at all**. The decision-curve conclusions of Section 8.3 are therefore the ones to distrust, by roughly a factor of two in the critical value, and the surface conclusions are the ones the calibration may already have covered. Neither statement is a coverage guarantee, and this version does not have one.

The safeguard we described as conservative — resolving only at the upper end of the critical value's Monte Carlo interval — moves coverage by less than a point, because it controls the Monte Carlo error in q and not the error in what q estimates. And no estimator among the five tested reaches nominal at this corpus's draw counts; the best, the empirical quantile at the upper end of its order-statistic interval, reaches 92.2% at 1.52 times the width. **The**

repair is more draws, not a better critical value: at the modal surface family coverage runs 82.0% at 33 draws and 95.8% at 400, and a draw refits the whole design space, which is why this version does not have them. We report the bands as computed and say what they are worth rather than substituting a wider estimator that is also wrong and merely less visibly so.

Some of what the decision-curve comparison counts is arithmetic. 841 cells of the corpus's curve families take the value zero in at least nine draws of ten, because at those thresholds both arms treat every case alike and the difference is zero by construction rather than by estimation. They pass the band's admission rule, which is stated in absolute units, and they inflate q_{emp}: excluding them moves the median ratio q_{emp}/q on those families from 2.09 to 1.80. The decision-curve family inherits every problem above undiluted, because the coverage calibration does not touch it.

Inference still rests on further assumptions. **The split point is fixed within a draw**, so split-position uncertainty is outside every interval in this paper. **The moving-block construction needs stationarity and mixing** that an event log is not guaranteed to have; neither is tested here. **The max-t band's family-wise coverage is asymptotic**, as is the multiplier bootstrap that estimates its critical value, and Section 10.4 shows what that asymptotics is worth at the 40 to 150 draws this corpus runs at. The block length is a convention, $\lceil n^{1/3} \rceil$; Section 10.3 varies it and the coverage of the reported interval does move with it, by up to 3.6% within a world, while the comparison the paper rests on does not.

The band's coverage is calibrated, and the calibration is itself an assumption. The inflation factor of Section 6.4 corrects the *scale* of the studentised statistic, which is the channel the (n, K) plane identifies. It does not correct a change in the shape of the maximum's distribution; it is estimated on a pointwise interval and applied to a simultaneous band; and it assumes the simulated worlds' dependence on K/n transfers. A reader who rejects the transfer should read the region labels and ρ as illustrative and rely on the objects that need no band: the point estimates, the decomposition, the all-cells sign-disagreement rate and the regret comparison.

The pipeline axis is crossed on one log only. Section 6.3 separates the model family from the encoding on the case study's log and finds the encoding axis the larger of the two. The other 15 pairs carry the two confounded, so their pipeline index remains a mixture, and whether the split found on one log transfers is not established here.

We do not measure reporting practice. The prevalence of one-number reporting in the published literature would be worth knowing and this paper does not establish it. The pilot that attempted it is in the archive with its own sampling error, coder error and resolution; it is a hypothesis, not evidence, and nothing here depends on it.

12. Conclusion

The incremental predictive performance of a recorded field is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, a decomposition of the surface's variance into the contribution of each choice, simultaneous confidence bands from a bootstrap that refits the model and whose critical value is calibrated for the register's cardinality, region labels that say when a sign is robust, and a decision-theoretic benchmark that measures what one number gets wrong.

On every pair, at least one admissible specification makes the register worth less than nothing. That is a statement about point estimates and needs no band; what the bands add is where a sign is *determined*, and of the 13 pairs that resolve any sign at all, 3 resolve both. Four further results are worth carrying away.

The choices move the answer, and no one of them is the culprit. Varying the baseline alone moves the increment by 0.073 AUC at the median log–target pair, but the largest first-order index at the median pair is 28.7%, the total higher-order share is 56.0%, and which axis leads is a property of the pair. An analyst cannot learn from somebody else's paper which of their own choices their answer turns on.

The observed sign depends jointly on the data, the target and the specification. On 18 of 19 pairs, between a tenth and nine tenths of the admissible cells are positive, and a conventional one-number report misstates the sign for a median 32.8% of the rest, and for 7.5% of the cells the corpus can resolve — a pooled ratio, driven by the pairs Section 5.4 names as construct-validity threats, and 3.5% on the 8 pairs whose register is a maintained one. Only one of the three arguments is conventionally reported.

There is a defensible collapse, and choosing between collapses is not where the value is. Lemma 1 identifies the decision-optimal one —

the weighted mean increment under a declared measure — but out of sample its advantage over the conventional report is small (0.00002 against 0.00450 AUC of excess regret), it reverses under rolling folds, and on the calibrated operational utility of Section 8.4 the rules mostly tie. What the decision-analytic reading does establish is sharper: on 40 of 118 models a threshold cannot be read as a cost ratio at all, and a desk that adopted this register on its *first-touch* number and deployed it at incident creation would be promised 1.71 true positives per thousand cases and receive 0.003.

The decision time is an argument of the estimand, not a preliminary. It fixes which cases are in scope, what is known about them, and which fields are admissible. On the case study, moving the prediction from first touch to incident creation costs −0.0297 AUC through the population, a further −0.0099 through the information once the population is held fixed, and −0.0076 more through the incidents that have no interaction record at all — before any baseline changes; and the same log's answer changes sign when the target changes, with 99.3% of the difference on the target and 0.0% on the cohort.

The practical recommendation is narrow and cheap once the surface exists: report the surface, the decomposition, the region and ρ; state the decision time and the target as parts of the estimand; make the absolute increments primary; use a ratio last, with a Fieller set, and only where its denominator is resolvably positive.

CRediT authorship contribution statement

Sudhir Dixit: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author has no affiliation with, and received no support from, any vendor of configuration management, IT service management or process mining software.

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Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used a large-language-model assistant, Anthropic Claude, for three purposes. The model identifier and the date of every session in which it was used are recorded in `AI-USE.md` in the archive, because a disclosure that names a product and not a version is not reproducible. The identifier for the final analysis and revision round is `claude-opus-5` (2026-08-24).

1. **Software.** The analysis scripts, the `fieldvalue` package and the verification harness were written with the assistant's help. Every result in this paper is produced by that code, the code is public, and `REPRODUCE.md` regenerates every table and figure from the raw data by one command.
2. **A use in a study that is in the archive and not in this paper.** A literature-prevalence pilot, described in Section 9.3 and reported nowhere in this paper or its supplement, was adjudicated by the author with the assistant's help from evidence dossiers extracted from full text. That is one machine-assisted adjudicator rather than two independent human raters, which is among the reasons the pilot carries no claim here. Its dossiers are in the archive; the model identifier used for that adjudication was not recorded at the time and is stated as unrecorded in `AI-USE.md`.
3. **Writing.** The assistant was used to draft and edit prose, which is a use beyond the language-and-readability editing an author may make without further comment, and is disclosed here for that reason. The author directed the structure and the argument, verified every claim against the result files, and takes full responsibility for the content of the publication.

No generative model produced, imputed, augmented or selected any datum, any result or any citation. The reference list was checked against the published records.

Data availability

All data are public. Event logs are obtained by DOI: BPI Challenge 2014 (https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35), BPI Challenge 2013 incidents (https://doi.org/10.4121/uuid:500573e6-accc-4b0c-9576-aa5468b10cee), and the further public logs whose DOIs and SHA-256 checksums are listed in `data/corpus/CHECKSUMS.txt` — 26 files are downloaded and checksummed in all, of which the registered role-assignment rules of Section 5 admit 13, yielding the 19 log–target pairs analysed here. The counts are different quantities and Table S5 accounts for every downloaded log that is not among the admitted pairs, in one of two ways: with the registered exclusion code the rules gave it, or, for the 1 logs downloaded as a held-out set for the out-of-corpus test, with the code HELD_OUT and the statement that they were never offered to the admission rules. The literature pilot's frame, sample, codes, adjudications and evidence dossiers are in the archive; the retrieved full texts are other publishers' copyrighted PDFs and are not redistributed, but `scripts/s06_audit2.py --fetch` re-retrieves them from the DOIs in `results/s06_sample.csv`. All derived results (`results/*.csv`), all figures and all code are in the archive below.

Code availability

Code, derived results, the pre-registered protocols, the amendment log and the verification harness are archived at the archived release cited in the data-availability statement (DOI reserved, inserted at proof) (release v20.0) and developed at https://github.com/sudhirdixit1/cmdb-routing-queue-baseline. The archive contains a `Dockerfile` and a `requirements.lock` pinning every transitive dependency; `REPRODUCE.md` gives the exact commands and the expected runtime.

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